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A
DO-IT-YOURSELF
INTRODUCTION TO NUMBER
THEORY

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Contents

Preface	iv
To the Instructor	v
To the Student	viii
1 The Fundamental Theorem of Arithmetic	1
1.1 The Fundamental Theorem in $\mathbb{Z}$	1
1.2 The Fundamental Theorem in the Gaussian Integers	6
1.3 An Integral Domain That Does Not “Enjoy” Unique Factorization	10
1.4 Polynomials Over a Field	11
2 An Overview of the Primes in $\mathbb{Z}$	16
2.1 More Arithmetic in $\mathbb{Z}$	16
2.2 Some Special Primes in $\mathbb{Z}$	18
3 Congruences	21
3.1 Congruences and the Ring $\mathbb{Z}_n$	21
3.2 The Euler ϕ -Function	24
3.3 Arithmetic Functions	29
3.4 Primitive Roots (mod p)	31
3.5 Communicating by Secret Code	33
4 Quadratic Reciprocity	35
4.1 Squares mod p	35
5 Sums of Two Squares and Pythagorean Triples	42
5.1 Which Positive Integers Are Sums of Two Squares?	42
5.2 Pythagorean Triples in $\mathbb{Z}$	45

Notes to the Instructor

50

Preface

This course, and its accompanying notes, were developed by (now retired) Professor James T. Cross over many years at the University of the South. The course has been consistently popular and it continues to be offered by friends of Prof. Cross, most recently by Profs. Drinen and Priestley. The notes were typeset in LaTeX in 2009 by Ye Katerina Totskaya and Van Nguyen. Subsequently, Chris Parrish has worked tirelessly to bring this work to completion. The notes are made available here by the Department of Mathematics and Computer Science as a tribute to Jim Cross's commitment to mathematics education.

To the Instructor

These notes on Number Theory were developed, class-tested, and modified several times, by James T. Cross over many years of teaching at the University of the South, Sewanee, Tennessee. Cross retired in the 1990s, but his notes have since continued to be used in Math 313 by other instructors.

Math 313 is a one-semester course intended for third-year undergraduate mathematics majors. It is usually comprised of six or eight students who have met the only prerequisite—a course in Discrete Structures, intended among other things to introduce them to proofs, and, in particular, to proofs using mathematical induction.

Typically, a daily classroom experience in Math 313 consists of two or three students each presenting their demonstrations of solutions (often proofs) to three or four exercises from the notes expressly assigned to them for that day. The other students—and the instructor, if needed—are encouraged to ask questions for clarification when necessary. During the course of the semester the students often have to build upon the work of their fellow students, as the exercises often require the results of previous exercises in order to be done easily and quickly.

If a bewildered student should come to the instructor the day before class to ask for help in presenting his/her assignment, there is usually a typical response that begins: Do you understand the results of the previous couple of exercises? Have you thought about how those results might make your assignment easier? And have you glanced ahead a little to see where all of this may be leading?

It takes some measure of maturity and patience on the part of students to take full advantage of such a response, and the major prerequisite for this course is a willingness to cultivate such maturity if it is not already present. The reward of such a regimen over the course of an entire semester, however, is often a marked improvement in the students' abilities to express themselves clearly for the benefit of their fellows. Also, in focussing upon clarity, the students may inadvertently learn to read mathematics with far more attention than before.

Cross's initial chapter begins by considering carefully how the simple division theorem and its systematic repetition in the Euclidean algorithm

lead to something with which all students are already familiar—the fundamental theorem of arithmetic in its original setting of the natural numbers. The notes then go on to show how an attempt to extend the notions involved to the Gaussian integers leads naturally to the ideas of groups, rings, ideals, integral domains, and isomorphisms—thus introducing the student to basic ideas of abstract algebra. Cross gives an example of a simple integral domain in which the fundamental theorem fails, and goes on to investigate how the theorem manifests itself in unsuspected places, such as in the fundamental theorem of algebra. Here it comes as something of a revelation to many students to see clearly how the splitting of an n -th degree polynomial over the complex numbers into n linear factors (primes) may be understood as nothing other than the familiar fundamental theorem of arithmetic as appropriately conceived in the integral domain $\mathbb{C}[x]$.

The titles of the remaining chapters and the titles of the sections therein indicate clearly how the course proceeds. Actually, we have never been able to work completely through all the notes in any one semester. Sometimes we choose to omit some sections, sometimes others. If a section is omitted, we simply outline and take for granted the main results of the section that are necessary to move forward. There has been at least one occasion, however, when the students became so enthusiastic about the enterprise that some of them agreed to meet together after the end of the semester in order to finish every detail on their own.

Usually, we have augmented the notes by making up and assigning weekly homework problems to be handed in by all students—sometimes including exercises from the notes themselves intended to reinforce ideas just introduced. And we usually have two or three hour tests and a final examination to ensure that definitions and particularly important proofs and procedures are remembered. The main focus of the class, however, has always been on the daily regimen and on the growth it fosters among students.

Although Cross later gives precise statements of the axioms for groups, rings, etc., he begins the notes by taking the point of view popularized long ago, that the natural numbers just are and that we have an intuitive understanding of their properties—their most subtle property being formalized by taking any one of three principles as fundamental and agreeing that the other two are consequences of it. Thus, the well-ordering principle, the principle of infinite descent, and the principle of mathematical induction should be already familiar—or readily grasped—by the students, who are expected to know how to employ such principles to establish results that hold for all natural numbers. A reminder of the precise statements of these principles follows this introduction.

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To the Student

I have arranged in this collection a sequence of do-it-yourself exercises that will introduce the student to the theory of numbers. These exercises are concerned with the Fundamental Theorem of Arithmetic in a variety of settings, the Quadratic Reciprocity Law, convolution of number-theoretic functions, sums of two squares, Pythagorean triples, etc. As anyone familiar with the subject matter will recognize, the study of these problems calls forth such topics as congruences, the Chinese Remainder Theorem, arithmetic functions, and other notions leading naturally to an introduction to groups, rings, integral domains, and fields.

It has been my practice to simply let the students do the exercises and present them in class. I have had very much fun in doing so, largely because the response from the students was so positive. Many of them not only did the exercises with pleasure, but also contributed to the ongoing revision of the collection. In particular I give thanks for Bob Kauffman, Mike Crowe, Jim Mathes, Charles Yeomans, Teresa Owen, Jay Woolfson, Emily Puckette, Lewis Jones, Nick Bennett, and Nowlin Randolph.

For a particularly good class, some of the exercises can be made strictly do-it-yourself and might not then be formally presented in class.

Have fun!

James T. Cross, January, 1995

THREE FAMOUS PRINCIPLES

The set of natural numbers is usually denoted by $\mathbb{N}$ or by $\mathbb{Z}^+$. The natural numbers are simply the familiar “whole numbers” $1, 2, 3, \dots$

(WOP) Well-ordering principle: Any nonempty set of $\mathbb{Z}^+$ contains a least element.

Restatement: If a set of natural numbers contains no least element, then the set is empty.

Re-restatement: If $P(n)$ is a proposition about the natural number n , and if the set of all n for which $P(n)$ is false contains no least element, then $P(n)$ is true for all natural numbers.

(PID) Principle of infinite descent: There is no strictly decreasing infi-

nite sequence of natural numbers. [You have to run into negative numbers, which are not in $\mathbb{Z}^+$, if you have a strictly decreasing infinite sequence of whole numbers.]

Restatement: Any assertion that implies there is a strictly decreasing sequence of natural numbers must be false.

Re-restatement: Any assertion whose negation implies there is a strictly decreasing sequence of natural numbers must be true.

(POMI) Principle of Mathematical Induction: If $P(n)$ is a proposition about the natural number n , if $P(1)$ is true, and if $P(n)$ implies $P(n+1)$, then $P(n)$ is true for all natural numbers n .

In the natural numbers, each of these three principles implies the other two! This means that if we accept any one of the three principles as a “self-evident axiom” then we can logically deduce the other two as “theorems” about $\mathbb{Z}^+$.

The question of which one of the three we take as an axiom (i.e., which one of the three we take as “most fundamental”) is really a question of aesthetics. The well-ordering principle is probably the most popular choice because it is stated in terms of sets and it can be stated very briefly—and speaking in terms of sets seems to be the preferred way of speaking of modern mathematics.

The principle of infinite descent is usually associated with Pierre de Fermat (1605?-1665) and the principle of mathematical induction with Blaise Pascal (1623-1662). This is not to say that they “discovered” the principles first, but that they were the ones who emphasized the fundamental roles in number theory played by the principles.

Finally, a note about some conventions used in these notes: Exercises are numbered, but definitions are not. Instead, definitions are introduced in bold-face, and most of the terms being defined re-appear in the index so that they can be found again.

Chapter 1

The Fundamental Theorem of Arithmetic

1.1 The Fundamental Theorem in $\mathbb{Z}$

Definition. Let $\mathbb{Z}$ denote the set of all integers (the counting numbers and their negatives together with 0) and let a , b , and c be in $\mathbb{Z}$: if $ab = c$, then each of a and b **divides** c (written $a|c$ and $b|c$) and c is a **multiple** of each of a and b . If u is in $\mathbb{Z}$ and $u|1$, then u is a **unit** in $\mathbb{Z}$.

Exercise 1.1. What are the units in $\mathbb{Z}$? Show that a unit in $\mathbb{Z}$ divides every member of $\mathbb{Z}$. Hint: The multiplicative unity 1 is of course a unit, but $\mathbb{Z}$ has a unit different from unity. What is it?

Exercise 1.2. If a and b are in $\mathbb{Z}$ and $a = ub$ for some unit u in $\mathbb{Z}$, then there is a unit v in $\mathbb{Z}$ such that $b = va$.

Definition. If a and b are in $\mathbb{Z}$ and $a = ub$ for some unit in $\mathbb{Z}$, then a and b are **associates**.

Exercise 1.3. If a and b are associates, then each divides the other. Conversely, if each divides the other, they are associates.

Exercise 1.4. If a is in $\mathbb{Z}$, what are a 's associates in $\mathbb{Z}$?

Exercise 1.5. If a is in $\mathbb{Z}^+$ (the positive integers), then there are integers, q and r , in $\mathbb{Z}$ with $r = 0$ or $r = 1$, such that $a = 2q + r$. (Hint: mathematical induction on a)

Exercise 1.6. If a is in $\mathbb{Z}^+$, then there exist q and r in $\mathbb{Z}$, with $r = 0, 1$, or 2 , and such that $a = 3q + r$.

Exercise 1.7. If a and b are in $\mathbb{Z}^+$, then there exist q and r in $\mathbb{Z}$, with $0 \leq r < b$, and such that $a = bq + r$.

Exercise 1.8. Find integers q and r such that $12 = 5q + r$, where $0 \leq r < 5$. Find integers q and r such that $5 = 12q + r$, where $0 \leq r < 12$. Find integers q and r such that $-12 = -5q + r$, where $0 \leq r < |-5|$.

Now you can see what we really want:

Exercise 1.9. The Divisor Theorem in $\mathbb{Z}$: If a and b are in $\mathbb{Z}$ and b is not 0, then there exist q and r in $\mathbb{Z}$, such that $a = bq + r$, where $0 \leq r < |b|$.

Be alert for the far-reaching consequences of this theorem.

Exercise 1.10. Give a and b various values and compare q and r . Draw some pictures on a number line. Don't be insulted because this seems to be a 4th grade exercise. It will help you be a better teacher of 4th graders, or of 14th graders.

Definition. Let a and b be in $\mathbb{Z}$ and suppose g is in $\mathbb{Z}$ and $g|a$ and $g|b$. Then g is a **common divisor** of a and b . If g is a common divisor of a and b and g has the property that every common divisor of a and b divides g , then g is a **greatest common divisor (GCD)** of a and b . We will denote a GCD of a and b by $\gcd(a, b)$. (You may be somewhat mystified by what seems to be an effort to make a simple concept appear more complex. Please be patient; you will see that this definition of a GCD will generalize readily to other mathematical entities, which in some cases we will call integers, and in which we don't have the handy ordering ($a < b$, etc.) that we have in $\mathbb{Z}$.)

Exercise 1.11. Find two GCD's of 12 and -38.

Exercise 1.12. If g is a GCD of a and b in $\mathbb{Z}$, then so is g 's associate in $\mathbb{Z}$, but there are no others. (If h is a $\gcd(a, b)$, then $h|g$ and $g|h$.)

Exercise 1.13. Let a and b be in $\mathbb{Z}$ and not both 0. Let $S = \{ax + by : x \text{ and } y \text{ are in } \mathbb{Z}\}$. S is closed under addition and under multiplication by members of $\mathbb{Z}$. That is, if s and s' are in S and z is in $\mathbb{Z}$, then $s + s'$ and zs are in S .

Exercise 1.14. (For those who have studied abstract algebra; we will revisit this exercise later for everyone.) The set S of Exercise 1.13 is an ideal of the ring $\mathbb{Z}$.

Exercise 1.15. Let a and b be -6 and 4, respectively, and $S = \{-6x + 4y : x \text{ and } y \text{ are in } \mathbb{Z}\}$. List enough members of S so that you can recognize a simpler way to define S . What is the least positive member of S ?

Exercise 1.16. If a and b are in $\mathbb{Z}$, not both 0, and $S = \{ax + by : x \text{ and } y \text{ are in } \mathbb{Z}\}$, then S contains a least positive member, d .

Exercise 1.17. (Use the notation of Exercise 1.16.) The member d divides both a and b . (Start with the Divisor Theorem. Remember that d is in S . Is a in S ?)

Exercise 1.18. (Use the notation of Exercise 1.16.) If z is in $\mathbb{Z}$ and $z|a$ and $z|b$, then $z|d$. (Again remember that d has the ticket of admission to S .)

Exercise 1.19. (Use the notation of Exercise 1.16.) Put Ex's 1.17 and 1.18 together and conclude that d is a GCD of a and b .

Exercise 1.20. (Use the notation of Exercise 1.16.) If z is in $\mathbb{Z}$, and $d|z$, then z is in S . If s is in S , then $d|s$. Thus $S = \{nd : n \text{ is in } \mathbb{Z}\}$.

Exercise 1.21. (Again for those who have studied abstract algebra, and again be assured that we will come to this again when everyone has the tools to participate.) The ideal S of Exercise 1.13 and 1.16 is a principal ideal, generated by d . This is not surprising, since every ideal in $\mathbb{Z}$ is principal. Can you prove it?

Exercise 1.22. If 1 is a GCD of a and b , what does the set $S = \{ax + by : x \text{ and } y \text{ are in } \mathbb{Z}\}$ look like? What if 2 is a GCD of a and b ? What if 25 is a GCD of a and b ?

Exercise 1.23. Suppose that you have a set of two-pan balances and an unlimited supply of 6-pound weights and 4-pound weights. Is it possible to weigh out exactly 15 pounds of sugar? (Argue by the use of the set S of the preceding exercises.) What if you have plenty of 25-pound weights and 46-pound weights?

We see that if a and b are in $\mathbb{Z}$, not both 0, then they have a GCD in $\mathbb{Z}$. (Of course, then they have two, according to Exercise 1.12.) Our proof was of the existence type; it doesn't give us a method by which we can chase down a GCD of a and b . There is an old **algorithm (Euclidean)** which enables one to do so. To illustrate the method, we find a GCD of 4827 and 32586.

$$\begin{array}{ll}
 32586 &= 4827(6) + 3624, & a &= bq_1 + r_1, \\
 4827 &= 3624(1) + 1203, & b &= r_1q_2 + r_2, \\
 3624 &= 1203(3) + 15, & r_1 &= r_2q_3 + r_3, \\
 1203 &= 15(80) + 3, & r_2 &= r_3q_4 + r_4, \\
 15 &= 3(5) + 0. & r_3 &= r_4q_5 + 0.
 \end{array}$$

The last nonzero remainder in this process (r_4 in this case) is a GCD of a and b . Why? The last line shows that r_4 divides r_3 . Then from the next-to-last line we see that r_4 divides r_2 . How do we see this? Now, keep climbing the column until you find that r_4 divides b and then a . Next, suppose d is a common divisor of a and b . The top line shows that d divides r_1 . Now, keep descending the column until you see that d divides r_4 .

Exercise 1.24. Describe the above algorithm in your own words and explain why it can not fail to identify a GCD of a and b . Why can the steps in the procedure not continue indefinitely?

Exercise 1.25. Use the Euclidean Algorithm to find a GCD for each of the following pairs of integers: 36 and 188; 36 and -188; 25 and 147; -389 and 12465.

Exercise 1.26. Explain why it is true that there exist integers x_0 and y_0 such that $32586x_0 + 4827y_0 = 3$. Then using (x_0, y_0) as a base point on the graph of $32586x + 4827y = 3$, write parametric equations of the line and show how to generate all (infinitely many) integer solutions of the equation. Also show that the same applies when 3 is replaced by any multiple of 3.

Exercise 1.27. Find integers x and y such that $32586x + 4827y = 3$. I'll help you get started. Go back to the display in which we found a GCD of the two integers:

$$\begin{aligned} 3 &= 1203 + 15(-80) \\ &= 1203 + [3624 + 1203(-3)](-80) = 3624(-80) + 1203(241) \\ &= 3624(-80) + [4827 + 3624(-1)](241) = 4827(241) + 3624(-321) \\ &= \dots \end{aligned}$$

Now you finish. (Then find all (infinitely many) pairs (x, y) of integers such that (x, y) is on the graph of the equation.) Put the algorithm in your own words and explain why it must succeed in expressing a GCD of two integers as a “linear combination” of the two integers.

Exercise 1.28. For each given pair of integers of Exercise 1.25 give your GCD as a linear combination of the two integers.

Definition. If 1 is a GCD of the two integers, a and b , then a and b are said to be **relatively prime** or **coprime** and each is said to be relatively prime to the other or coprime with the other.

Exercise 1.29. The integers a and b are relatively prime if and only if 1 and -1 are their only common divisors. This is true if and only if there exist integers x and y such that $ax + by = 1$.

Exercise 1.30. If each of c , a , and b is in $\mathbb{Z}$ and $c|ab$ and c is relatively prime to a , then $c|b$. Hint: $cx + ay = 1$. Multiply through by b .

Definition. Let n be in $\mathbb{Z}$ and n not be 0 and n not be a unit. If every divisor of n is either a unit or an associate of n (thus the only divisors of n are 1, -1 , n , and $-n$), then n is a **prime** in $\mathbb{Z}$; if n is not prime in $\mathbb{Z}$ (thus n has a divisor other than 1, -1 , n , $-n$), then n is said to be **composite**.

Exercise 1.31. The set $\mathbb{Z}$ of integers can be partitioned into four non-intersecting classes. These are 0, the units, the primes, and the composites.

Exercise 1.32. Find all the primes between 0 and 100 and all those between -100 and 0.

Exercise 1.33. If p is a prime in $\mathbb{Z}$ and $p|ab$, then $p|a$ or $p|b$. If n is composite in $\mathbb{Z}$, n may divide ab without dividing a or b . Give examples.

Exercise 1.34. Show that $3^{100} \neq 7^k$ for any integer k . (Hint: 3 is a prime dividing the left side of the inequality. Use Exercise 1.33).

Exercise 1.35. If p and q are primes in $\mathbb{Z}$ and $p|q$, then $p = q$ or $p = -q$.

Exercise 1.36. Let a be composite in $\mathbb{Z}$. Then $a = bc$ for some b and c in $\mathbb{Z}$, where neither b nor c is a unit or an associate of a ; if a is positive, then there exist b and c in $\mathbb{Z}$ with $1 < b < a$ and $1 < c < a$ and such that $a = bc$.

Exercise 1.37. If a is in $\mathbb{Z}$ and $a > 1$, then a is a product of positive primes. (We consider a single prime to be a product having one factor.) (Hint: Suppose for some bad $a > 1$, a is not a product of positive primes. Let L be the least such bad one. Is L composite? Proceed.)

Exercise 1.38. If a is in $\mathbb{Z}$ and a is not 0 and a is not a unit, then a is a product of primes.

Exercise 1.39. Express each of the following integers as a product of primes: 48, -48 , 3624, -3624 , 10000, -10000 . Do you have some choice in each case? Exactly what choices do you have?

Exercise 1.40. Check that $36 = 2^2 3^2 = (-2)^2 (-3)^2 = 2(-2)3(-3)$. Show that any prime factor of 36 divides 2 or 3 and hence is 2 or -2 or 3 or -3 .

Exercise 1.41. Show that 36 can be factored into positive primes in one and only one way: $36 = 2^2 3^2$. (Hint: Suppose that $2^2 3^2 = p_1 p_2 p_3 \dots p_r$, where the p 's are positive primes, not necessarily distinct. Does 2 divide one of them? Is 2 one of them? Cancel 2 from both sides. Proceed.)

Exercise 1.42. If a is in $\mathbb{Z}$ and $a > 1$, then a can be factored into positive primes in one and only one way.

Exercise 1.43. Suppose n is in $\mathbb{Z}$ and $n \neq 0$ and n is not a unit. Then a factorization of n into primes is essentially unique. That is, if we have two factorizations, F_1 and F_2 , of n into primes, and the prime p appears k times as a factor in F_1 while its associate (negative) appears m times, so that the total number of times that p or $-p$ occurs in F_1 is $k + m = r$, then r is also precisely the number of times that p or $-p$ occurs in F_2 .

Now put Ex.'s 1.38 and 1.43 together to establish the **Fundamental Theorem of Arithmetic** in $\mathbb{Z}$:

Exercise 1.44. If a is in $\mathbb{Z}$ and $a \neq 0$ and a is not a unit, then a can be factored into primes, and if the distinction between a prime and its associate is ignored, then the factorization is unique.

You should note that it is the Divisor Theorem that enabled us to waltz straight to the Fundamental Theorem.

Exercise 1.45. Produce a road map from Exercise 1.9 to Exercise 1.44.

One is inclined to think that this emphasis on the Fundamental Theorem is a lot of fuss about nothing since the theorem merely confirms what was

learned in elementary school. In order to help you gain some perspective about this point and also to develop some tools with which to solve some simple-sounding and tantalizing problems (for example, what positive integers are sums of two squares?), we are now going to examine some sets whose members share so many properties with $\mathbb{Z}$ that we call them integers (of course we will have to employ adjectives to distinguish them from $\mathbb{Z}$). We will see that in some of these sets of “integers,” factorization into primes is unique, while in others it is not, and we will see later that this uniqueness property is a powerful tool in solving some of those tantalizing problems.

1.2 The Fundamental Theorem in the Gaussian Integers

Definition. The subset $G = \{x + yi : x \text{ and } y \text{ are in } \mathbb{Z}\}$ of the complex numbers is called the set of **Gaussian Integers**.

Exercise 1.46. Plot the set of Gaussian Integers in the complex plane.

Definition. A **group** is a nonempty set, S , together with a binary operation on S (which we will denote at this point by the symbol, $+$, although the operation may very well not be the common garden variety of addition), which has these four properties:

1. If a and b are in S , then so is $a + b$. (S is closed relative to the operation.)
2. If a , b , and c are in S , then $a + (b + c) = (a + b) + c$. (The operation is associative.)
3. There is a member, e , in S , such that $a + e = e + a = a$. (There is an identity member for the operation.)
4. If a is in S , there is a member, b , in S such that $a + b = e$. (Every member of S has an inverse in S .)

If a group has the additional property that $a + b = b + a$ for all a and b in the group, then the group is said to be **commutative**, or **abelian**.

Exercise 1.47. G is an abelian group relative to addition. So is $\mathbb{Z}$.

Definition. A **ring** R is a nonempty set with two operations (which we will call addition and multiplication, although again they may not be the usual sort), which have these three properties:

1. R is an abelian group relative to addition.
2. Multiplication in R is associative; that is $a(bc) = (ab)c$.
3. Multiplication distributes over addition; $a(b + c) = ab + ac$.

If a ring has the property that multiplication is commutative, then it is called a **commutative ring**. If it has an identity for multiplication (distinct from its identity for addition), then it is a **ring with unity**.

(You may be acquainted with one ring which is not commutative: the ring of $n \times n$ matrices.)

Exercise 1.48. G is a commutative ring with unity. So is $\mathbb{Z}$.

Definition. Let $a = x + yi$ be a complex number, where x and y are real numbers. The **norm**, $N(a)$, of a is $x^2 + y^2$. The **complex conjugate** $\bar{a}$, of a , is $x - iy$.

Exercise 1.49. The norm of a is the product of a and its complex conjugate: $N(a) = a\bar{a}$. If $a = x + yi$, then $N(a)$ is the square of the distance from $(0,0)$ to (x,y) in the complex plane. If a and b are complex numbers, then $N(ab) = N(a)N(b)$. If a is in G , the norm of a is a nonnegative member of $\mathbb{Z}$. If a and b are in G and a divides b in G , then $N(a)$ divides $N(b)$ in $\mathbb{Z}$.

Exercise 1.50. The ring G is an **integral domain**; that is, a commutative ring with unity in which the product of any two nonzero members of G is nonzero. The same is true of $\mathbb{Z}$. (One usually thinks of $\mathbb{Z}$ as the model for an integral domain.)

Now we are going to do some exercises to convince you that the Fundamental Theorem of Arithmetic is valid in G . (I promise you that you will see an integral domain in which the Fundamental Theorem is *not* valid.) You would find it rewarding to try to navigate your own way to the goal by attempting to adapt Exercises 1.1–1.44 to G , peeking at our program only when (and if) you must.

Exercise 1.51. Look at the definition of a unit in $\mathbb{Z}$ and what it means to say that a divides c in $\mathbb{Z}$. Now, of course you can define a unit in G and say what it means that a divides c in G . Answer: the member u of G is a unit if there is a member v of G such that $uv = 1$. To say that a divides c in G (written $a|c$) means that there is some b in G such that $ab = c$. Thus the units in G (or in $\mathbb{Z}$) are the divisors of 1.

Exercise 1.52. If u is a unit in G , then $N(u) = 1$. There are exactly four members of G having norm 1 and each of these four is a unit. The set U of units in G is $\{1, i, -1, -i\}$. This set U is a group relative to multiplication.

(We may as well note that U is the group of 4^{th} roots of 1. If n is a positive integer, there are n n^{th} roots of 1, and they are a group relative to complex number multiplication.)

Exercise 1.53. Now you can say what it means that a and b are associates in G and show that if a and b are associates, then they have the same norm. Is the converse true: if a and b have the same norm, then they are associates?

Exercise 1.54. Let a and b be associates in G . Describe their relative positions as points in the complex plane.

Exercise 1.55. Divide $5 + 6i$ by $3 + i$ to obtain

$$\frac{5 + 6i}{3 + i} = \frac{21 + 13i}{10} = 2 + i + \frac{1}{10} + \frac{3}{10}i.$$

Plot $\frac{21+13i}{10}$ in the complex plane. Also plot $2 + i$ and $\frac{1}{10} + \frac{3}{10}i$.

Exercise 1.56. Show that $5 + 6i = (3 + i)(2 + i) + i$ and verify that there exist q and r in G such that $5 + 6i = (3 + i)q + r$, where $0 \leq N(r) < N(3 + i)$.

Exercise 1.57. If c is a complex number, then there is a Gaussian Integer q such that $N(c - q) < 1$. If a and b are complex numbers and $b \neq 0$, then there is a Gaussian Integer q such that $N((a/b) - q) < 1$.

Exercise 1.58. If each of a , b , and q is a Gaussian Integer and r is a complex number and $a = bq + r$, then r is a Gaussian Integer.

Exercise 1.59. If each of a and b is in G and $b \neq 0$, then there exist q and r in G with $r = 0$ or $N(r) < N(b)$ and such that $a = bq + r$. (This result is, of course, the **Divisor Theorem** in G .)

Exercise 1.60. Practice a bit. Take some a 's and b 's and find q 's and r 's. Then you will really understand "what's going on".

Definition. If a , b , and g are in G and $g|a$ and $g|b$, then of course we say that g is a common divisor of a and b . If g is a common divisor of a and b and every common divisor of a and b divides g , then again of course we say that g is a **greatest common divisor (GCD)** of a and b . (Now your patience is paying off; you can see why we defined a gcd in $\mathbb{Z}$ as we did.)

Exercise 1.61. If g is a GCD of a and b in G and u is a unit in G , then ug is a GCD of a and b in G .

Exercise 1.62. Let each of a and b be in G and not both 0. Let $S = \{ax + by : x \text{ and } y \text{ are in } G\}$. Then S is closed under addition and under multiplication by members of G , S contains a member d with least positive norm, d divides both a and b in G and any common divisor of a and b in G divides d in G , d is a GCD of a and b in G , and $S = \{nd : n \text{ is in } G\}$.

Exercise 1.63. Every pair of Gaussian integers, at least one of which is nonzero, has exactly four GCD's.

Definition. If T is a group and S is a subset of T that is itself a group relative to the operation that makes T a group, then S is a **subgroup** of T .

Definition. If S is a subgroup of the additive group of a ring R and S has the property that for any s in S and any r in R , rs and sr are in S , then S is an **ideal** of R .

Exercise 1.64. If R is a commutative ring with unity and a is a member of R , then the set $\{ra : r \text{ is in } R\}$ is an ideal of R , called the **principal ideal** generated by a .

Exercise 1.65. The set S of Exercise 1.13 and the set S of Exercise 1.62 are principal ideals of $\mathbb{Z}$ and G , respectively.

Exercise 1.66. Every ideal of $\mathbb{Z}$ and every ideal of G is a principal ideal.

Exercise 1.67. The Euclidean Algorithm can be used to chase down GCD's in G . For example, find a GCD of $4 + 10i$ and $1 + 5i$ by means of the Algorithm. (Start by dividing $4 + 10i$ by $1 + 5i$, getting a quotient and a remainder, where the remainder has norm less than that of $1 + 5i$. Compare with Exercise 1.25.) Answer: $1 + i$ (or any associate of $1 + i$).

Exercise 1.68. Find Gaussian integers, x and y , such that $(1 + 5i)x + (4 + 10i)y = 1 + i$. (Compare with Exercise 1.27.)

Exercise 1.69. For each of the following pairs of Gaussian integers, find a GCD. Then write the GCD as a linear combination of the two given integers: $3 + i$ and $1 + 2i$; $1 + 18i$ and $11 + 13i$; 36 and 188.

Definition. If 1 is a GCD of two Gaussian integers, then of course they are said to be **relatively prime** or **coprime** in G and each is said to be relatively prime to the other.

Exercise 1.70. The Gaussian integers a and b are relatively prime if and only if their common divisors are 1, i , -1 , $-i$. This is true if and only if there exist Gaussian integers, x and y , such that $ax + by = 1$.

Exercise 1.71. If a and b are in $\mathbb{Z}$ and are relatively prime in $\mathbb{Z}$, then they are relatively prime in G . (There is more to this than meets the eye. Is it not reasonable to think that even though a and b have no nontrivial common divisor in $\mathbb{Z}$, they might have one in G ?)

Exercise 1.72. If each of c , a , and b is a Gaussian integer and c and a are relatively prime and c divides ab , then c divides b .

Definition. Let g be in G and $g \neq 0$ and g not a unit. If every divisor of g is either a unit or an associate of g , then g is **prime** in G ; if g is not prime in G (thus g has a divisor different from g , ig , $-g$, or $-ig$), then g is **composite**.

Exercise 1.73. The Gaussian integers can be partitioned into four nonintersecting classes: 0, the units, the primes, and the composites.

Exercise 1.74. The numbers $1 + i$, $1 + 2i$, 3, 7, 11, and 19 are all prime in G . The numbers $1 + 5i$, 2, 5, and 13 are all composite in G . (Hint: Suppose $1 + i = ab$, where neither a nor b is a unit. Then $N(1 + i) = 2 = N(a)N(b)$, etc.)

Exercise 1.75. Let p be a prime in G and let a and b be in G . If $p|ab$, then $p|a$ or $p|b$.

Exercise 1.76. If p and q are prime in G and $p|q$, then $p = q$ or p is an associate of q . That is, $p = q$ or $p = qi$ or $p = -q$ or $p = -qi$.

Exercise 1.77. Fundamental Theorem of Arithmetic in G : If g is in G and $g \neq 0$ and g is not a unit, then g is a product of primes. (Hint: Suppose there is some bad g (one that is not a product of primes). Then let b be a bad one with least norm. Then what?) Moreover, if we ignore the distinction between a prime and its associates, then the factorization is unique.

You can see that gcd's are related to unique factorization. Ideals were conceived by E. Kummer in the 1800's as "ideal numbers" (Was $\sqrt{-1}$ a sort of ideal number at one time? Indeed, would -1 have been considered "ideal" when first conceived?) Kummer's ideals were used in order to provide ideal gcd's of numbers in certain domains in which it isn't true that every pair of members, not both 0, have a gcd. (This is a simplification but maybe you can get the idea: $\{6x + 4y : x \text{ and } y \text{ are in } \mathbb{Z}\} = \{2x : x \text{ is in } \mathbb{Z}\}$. Now, suppose there were no number to play the role of 2 here. You could still talk about the ideal $\{6x + 4y : x \text{ and } y \text{ are in } \mathbb{Z}\}$ and maybe you could arrange things so as to think of the ideal itself as a gcd of 4 and 6.) It was a valiant attempt to prove **Fermat's Last Theorem**, which would have been provable at the time if there weren't some domains which do not "enjoy" unique factorization into primes. We will have more to say about Fermat's Last Theorem later, but we should state it here: if x, y, z , and n are positive integers and $n > 2$, then $x^n + y^n \neq z^n$.

I hope that you have had fun with this new integral domain, the Gaussian Integers, and that you were particularly impressed by the kinship between G and $\mathbb{Z}$. Have you thought about how to identify the primes in G ? We can not address that problem efficiently until we return to our study of $\mathbb{Z}$ and develop some helpful tools. We shall do that presently. At this point I think you deserve to see an integral domain in which the Fundamental Theorem is not valid and a familiar one in which it is.

1.3 An Integral Domain That Does Not "Enjoy" Unique Factorization

The word "enjoy" is not my concoction; it is commonly used in this context to ascribe emotion to an abstract concept.

In the complex numbers, let J denote the set, $\{x + y\sqrt{3}i : x \text{ and } y \text{ are in } \mathbb{Z}\}$.¹

Exercise 1.78. Plot J in the complex plane.

Exercise 1.79. J is an integral domain containing $\mathbb{Z}$.

Exercise 1.80. If a is in J , then $N(a)$ is a nonnegative member of $\mathbb{Z}$. If a and b are in J and $a|b$ in J , then $N(a)|N(b)$ in $\mathbb{Z}$.

Exercise 1.81. The units (divisors of 1, of course) in J are the set $\mathbb{U}$, each of whose members has norm 1; $\mathbb{U} = \{1, -1\}$.

Exercise 1.82. If a and b are in J and $ab = 2$, then one of a and b is a unit. Thus, 2 is prime in J . So is -2 . (You can make your own definition of a prime in J .)

Exercise 1.83. If a and b are in J and $ab = 1 + \sqrt{3}i$, then one of a and b is a unit. Thus $1 + \sqrt{3}i$ is prime in J . So is $1 - \sqrt{3}i$.

Exercise 1.84. The primes 2 and $1 + \sqrt{3}i$ are not associates. Neither are 2 and $1 - \sqrt{3}i$. (Remember that if a and b are associates, then one is a unit times the other.)

Exercise 1.85. The member 4 of J factors into primes in two distinct ways:

$$4 = 2 \times 2 = (1 + \sqrt{3}i)(1 - \sqrt{3}i).$$

The Fundamental Theorem of Arithmetic is not valid in J . The domain J can be enlarged to obtain an interesting and useful integral domain in which the Fundamental Theorem is valid. That domain can be used to show that there exists no triple $\{x, y, z\}$ of positive integers such that $x^3 + y^3 = z^3$. This is a special case of Fermat's Last Theorem mentioned earlier: if n is in $\mathbb{Z}$ and $n > 2$, there is no triple of positive integers such that $x^n + y^n = z^n$.

1.4 Polynomials Over a Field

Now we are going to look at an integral domain with which you are familiar, almost as familiar as you are with $\mathbb{Z}$. You will see it from a new perspective and be impressed by its kinship with $\mathbb{Z}$ and with G . At points in our discussion we will need the definition of a field:

Definition. A **field** is an integral domain in which all nonzero members are units.

Examples: The rational numbers, $\mathbb{Q}$; the real numbers, $\mathbb{R}$; the complex numbers, $\mathbb{C}$; the set $\{a + bi : \text{each of } a \text{ and } b \text{ is in } \mathbb{Q}\}$. (You should check out that what I've said is true.) These are all examples of infinite fields; we will see later that there are some interesting and useful finite fields with which you probably are not yet familiar. ²

Definition. Let D be a ring (not necessarily a field). A **polynomial** (in one variable) over D is an expression of this type:

$$a_0 + a_1x + a_2x^2 + a_3x^3 + \dots + a_nx^n,$$

where the a 's are members of D and n is a nonnegative integer. The set of all such polynomials will be denoted by $D[x]$.

Of course you have seen these polynomials since you studied elementary algebra.

Exercise 1.86. Determine whether each of the following polynomials is a member of $\mathbb{C}[x]$, $\mathbb{R}[x]$, $\mathbb{Q}[x]$, $G[x]$, and $\mathbb{Z}[x]$:

- $3 + 2x + 4x^2$;
- $1/2 + 3x$;
- $(1 + 2i)x$;
- 5 ;
- $3 + \pi x^3$;
- $\begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix} + \begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix} x^4$.

Definition. Let $P(x)$ denote a polynomial in $D[x]$, where D is a ring. If $P(x) \neq 0$ (that is, some $a_i \neq 0$) and $P(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \cdots + a_nx^n$, where $a_n \neq 0$, then n is the **degree** of $P(x)$. If $P(x) = 0$, then $P(x)$ has no degree.

Exercise 1.87. Give a polynomial over $\mathbb{Q}$ having degree 2, give one having degree 1, give one having degree 0, and give one having no degree.

Exercise 1.88. Let D be a ring. The polynomials over D having degree 0 are precisely the nonzero members of D .

We could make formal definitions of the sum and the product of two polynomials in $D[x]$. Let us not do so. You have added and multiplied polynomials enough so that we can avoid this bit of formalism. (Notice that we didn't define addition and multiplication in $\mathbb{Z}$, either.) Again, I think it is unnecessary for me to tell you what it means that $a(x)$ divides $b(x)$ in $D[x]$. What does it mean? If polynomials are added or multiplied, what can you say about the degree of the resulting sum or product?

Exercise 1.89. If D is an integral domain (remember, every field is an integral domain), then so is $D[x]$, and the units of $D[x]$ (divisors of 1, of course, where 1 denotes the unity member of D and of $D[x]$) are precisely the units of D .

Exercise 1.90. Let $P(x)$ and $P_1(x)$ be in $D[x]$, where D is an integral domain. What would it mean to say that they are associates in $D[x]$?

Exercise 1.91. Let $P(x) = 1/2 + 3x + 4x^2 + 2/3x^4$ and $P_1(x) = 3 + 18x + 24x^2 + 4x^4$ be polynomials in $\mathbb{Q}[x]$. Then $P(x)$ does not belong to $\mathbb{Z}[x]$, but $P(x)$ and $P_1(x)$ are associates in $\mathbb{Q}[x]$.

Exercise 1.92. Let $1/2 + 3x + 4x^2 + 2/3x^4$ and $1 + 2x^2$ be polynomials in $\mathbb{Q}[x]$. Find polynomials $q(x)$ and $r(x)$ in $\mathbb{Q}[x]$ with $r(x) = 0$ or $\text{degree}(r(x)) < 2$ and such that $1/2 + 3x + 4x^2 + 2/3x^4 = (1 + 2x^2)q(x) + r(x)$.

Exercise 1.93. Let F be a field and let $a(x)$ and $b(x)$ be polynomials over F and let $b(x) \neq 0$. Then there exist polynomials $q(x)$ and $r(x)$ in $F[x]$ with $r(x) = 0$ or $\text{degree}(r(x)) < \text{degree}(b(x))$ and such that $a(x) = b(x)q(x) + r(x)$. (This result is, of course, the **Divisor Theorem** in $F[x]$.)

Exercise 1.94. Let F denote a field. Make up your own program that leads to the **Fundamental Theorem of Arithmetic** in $F[x]$.

Exercise 1.95. Take the two given polynomials of Exercise 1.92 and chase down a GCD by the Euclidean Algorithm. Do the same for $1 + 2x^2$ and $P_1(x)$, where $P_1(x)$ was given in Exercise 1.91. Do you get the same answer? How many such GCD's are there in each case? Are they all associates? In each case write your GCD as a linear combination of the two given polynomials.

Exercise 1.96. Give examples of irreducible polynomials (primes) in $\mathbb{Q}[x]$, in $\mathbb{R}[x]$, and in $\mathbb{C}[x]$.

In Chapter 5 you will see how to identify all primes in G , using those in $\mathbb{Z}$. You are pretty familiar with primes in $\mathbb{Z}$, although you will learn more about them in Chapter 2. I think that this is probably the point to teach you some significant facts about primes in $\mathbb{Q}[x]$, $\mathbb{R}[x]$, and $\mathbb{C}[x]$. First, we need a theorem:

Exercise 1.97. Let F be a field, $f(x)$ be in $F[x]$, and a in F . Use the divisor theorem in $F[x]$ to prove that $f(a) = 0$ (a is a zero of f) if and only if $(x - a) \mid f(x)$ in $F[x]$.

Now I am going to state the **Fundamental Theorem of Algebra**, which is easy to state and understand, but the proof of which does not belong in this collection but in a course in complex variables: let $f(x)$ be in $\mathbb{C}[x]$ and have degree one or more. (You must keep aware that if $f(x)$ is in $\mathbb{Q}[x]$ or $\mathbb{R}[x]$, then f is in $\mathbb{C}[x]$.) Then there is a member c in $\mathbb{C}$ such that $f(c) = 0$ (that is, f has a zero in $\mathbb{C}$). There is jargon for this: $\mathbb{C}$ is an **algebraically closed field**; you don't have to go to a bigger field in order to get a zero of a polynomial in $\mathbb{C}[x]$. (This does not mean, of course, that you can easily find the zero c .)

Exercise 1.98. Factor $f(x) = x^3 - 1$ into primes in $G[x]$, $\mathbb{R}[x]$, and $\mathbb{C}[x]$. Note that $x^3 - 1 = (x - 1)g(x)$ and the Fundamental Theorem of Algebra assures that g has a zero in $\mathbb{C}$.³ (The zero might be in $\mathbb{R}$ or even $\mathbb{Q}$.)

Exercise 1.99. The Fundamental Theorem of Algebra and Exercise 1.97 can be used repeatedly to imply that if $f(x)$ is in $\mathbb{C}[x]$ and has positive degree,

then f factors into n linear factors (possibly times a unit) in $\mathbb{C}[x]$; some of the factors may be repeated.

Exercise 1.100. The primes in $\mathbb{C}[x]$ are precisely the linear members. Don't let this slip by you; it is significant.

Now we turn to $\mathbb{R}[x]$, which is more interesting. Consider the mapping (function) g from $\mathbb{C}$ to $\mathbb{C} : g(c) = \bar{c}$. (The function takes a complex number (reals and rationals included, of course) to its complex conjugate.)

Exercise 1.101. If c and d are in $\mathbb{C}$, then $g(c+d) = g(c) + g(d)$ and $g(cd) = g(c)g(d)$. Moreover, g is a bijection from $\mathbb{C}$ to $\mathbb{C}$.

Exercise 1.102. If a is a real number, then $g(a) = a$. In particular, $g(0) = 0$.

Exercise 1.103. If a is a (complex) zero of the polynomial

$$r_0 + r_1x + r_2x^2 + \cdots + r_nx^n$$

in $\mathbb{R}[x]$, then so is $\bar{a}$. (What does it mean to say that a is a zero of f ?)

Exercise 1.104. Let r , s , and t be real numbers and a be a complex (maybe real, maybe not). Then $g(r + sa + ta^2) = r + s\bar{a} + t\bar{a}^2$.

Exercise 1.105. If a and $\bar{a}$ are zeros of $f(x)$ in $\mathbb{R}[x]$, where a is not real, then the polynomial $(x - a)(x - \bar{a}) = x^2 - (a + \bar{a})x + a\bar{a}$ is prime in $\mathbb{R}[x]$ and is a factor of $f(x)$ there.

Exercise 1.106. The primes in $\mathbb{R}[x]$ are linear or quadratic polynomials, the quadratic ones having conjugate pairs of complex nonreal zeros.

We see that in $\mathbb{C}[x]$ there is no prime with degree greater than 1 and that in $\mathbb{R}[x]$ there is no prime with degree greater than 2. Now we wonder about $\mathbb{Q}[x]$.

Exercise 1.107. Find a prime polynomial with degree 1, one with degree 2, one with degree 3, and one with degree 4 in $\mathbb{Q}[x]$.

There exist prime polynomials of all positive degrees in $\mathbb{Q}[x]$; it would take us too far afield (pun intended) to prove it, but we could surely do it. It is generally difficult to determine whether a given polynomial in $\mathbb{Q}[x]$ is prime there; there exist some tests which can be used in special cases.

It is not at all difficult, however, to determine whether a polynomial in $\mathbb{Q}[x]$ has a linear factor in $\mathbb{Q}[x]$.

Exercise 1.108. Let $f(x) = 2/3x^3 - 1/3x^2 - 1/3x - 1$ and $g(x) = 2x^3 - x^2 - x - 3$ be in $\mathbb{Q}[x]$. Show that f and g have the same zeros in $\mathbb{C}$ (and therefore, in $\mathbb{R}$ or $\mathbb{Q}$). This is not a big deal; f and g are associates in $\mathbb{Q}[x]$.

Exercise 1.109. Let $f(x)$ be in $\mathbb{Q}[x]$. Then by multiplying by an appropriate unit (rational number), one can find $g(x)$ in $\mathbb{Z}[x]$ that has the same zeros as $f(x)$.

Exercise 1.110. Suppose a/b (a and b in $\mathbb{Z}$ and $\gcd(a, b) = 1$) is a rational number that is a zero of the polynomial $g(x)$ in Exercise 1.108. Substitute a/b for x , set equal to 0, and show that $a|3$ and $b|2$, so that the only possible rational zeros of $g(x)$ are $1, -1, 3, -3, 1/2, -1/2, 3/2$ and $-3/2$.

Exercise 1.111. Find all (complex) zeros of the polynomial $f(x)$ of Exercise 1.108. Factor f in $\mathbb{Q}[x]$, $\mathbb{R}[x]$, and $\mathbb{C}[x]$.

Exercise 1.112. Generalize Exercises 1.108–1.110 and show how to find all rational zeros of a member of $\mathbb{Q}[x]$. (You will have proved the **Rational Root Theorem**.)

Exercise 1.113. Apply the **Rational Root Theorem** to the equation $x^2 - c = 0$, where c is a positive integer. Deduce that the square root of c is irrational unless c is a perfect square.

Chapter 2

An Overview of the Primes in $\mathbb{Z}$

2.1 More Arithmetic in $\mathbb{Z}$

Exercise 2.1. Let a and b be relatively prime positive members of $\mathbb{Z}$.

Show that $\sqrt{7} \neq a/b$. (Hint: Exercise 1.33) Then show that $\sqrt{7} \neq a/b$ for any members, a and b , of $\mathbb{Z}$. (Alternatively, use Exercise 1.113.)

Exercise 2.2. Let p be a positive prime and n an integer greater than 1. Then $(p)^{1/n} \neq a/b$ for members, a and b , of $\mathbb{Z}$.⁴ That is, $(p)^{1/n}$ is not a rational number.

Exercise 2.3. Do the prime factorizations of two members of $\mathbb{Z}$ provide a means of writing down a GCD? For example give a GCD of the two members, $2^3 3^5 17^2$ and $3^2 5 \cdot 17^2 37$. Put your method into words.

Definition. Let a and b be nonzero members of $\mathbb{Z}$. If m is in $\mathbb{Z}$ and each of a and b divides m , then m is a **common multiple** of a and b . If m is a common multiple of a and b and m divides every common multiple of a and b , then m is a **least common multiple** of a and b .

Exercise 2.4. Check that 6 is a least common multiple of 2 and 3, that 12 is a least common multiple of 4 and 6, and that -12 is a least common multiple of 4 and 6.

Exercise 2.5. If m is a least common multiple of a and b in $\mathbb{Z}$, then so is $-m$. If m and n are both least common multiples of a and b , then each divides the other and hence they are associates.

Exercise 2.6. If a and b are nonzero members of $\mathbb{Z}$ and d denotes $\gcd(a, b)$ and $a = a_1 d$ and $b = b_1 d$, then a_1 and b_1 are relatively prime.

Exercise 2.7. If a and b are nonzero members of $\mathbb{Z}$ and L denotes $\frac{ab}{\gcd(a,b)}$, then L is a least common multiple of a and b .

Exercise 2.8. Find a least common multiple of the two integers of Exercise 2.3. Put into words your method of doing so.

Is it obvious to you that there should be infinitely many primes in $\mathbb{Z}$? If so, you should be able to give some reasons. (It isn't obvious to me.) You can prove it though:

Exercise 2.9. Let n be a positive integer and suppose that for each positive integer i such that $1 \leq i \leq n$, P_i is a positive prime in $\mathbb{Z}$. Let $M = (P_1 P_2 P_3 \dots P_n) + 1$. Then M has a prime divisor, which cannot be P_i for any i such that $1 \leq i \leq n$.⁵

Exercise 2.10. Exercise 2.9 implies, à la Euclid, that $\mathbb{Z}$ contains infinitely many primes.

Exercise 2.11. Check that $2 + 1$, $(2)(3) + 1$, $(2)(3)(5) + 1$, and $(2)(3)(5)(7) + 1$ are all primes.

Exercise 2.12. Make a conjecture based on Exercise 2.11. Do a bit of checking on your conjecture.⁶

Now that you know that there are infinitely many primes, would you believe that there are arbitrarily long gaps between consecutive primes? That is, let your friends choose a positive integer n , no matter how large. Then you can prove that there exists a string of n consecutive composite positive integers. First we look at a special case:

Exercise 2.13. The 100 consecutive numbers in the set, $\{(101)! + 2, (101)! + 3, (101)! + 4, \dots, (101)! + 101\}$, are all composite.

Now we generalize:

Exercise 2.14. Let n be a positive integer. There exists a set of n consecutive composite positive integers.

Exercise 2.15. On the other hand it has been conjectured (but not proved) that there are infinitely many pairs of **twin primes** (primes, p and q , such that $q = p + 2$). Find a pair of twin primes, each member of which is greater than 100.

Exercise 2.16. You can sense that the primes are scattered very irregularly among the positive integers. You may therefore be surprised to discover that there is order in the chaos. Let x denote a positive number and $\pi(x)$ denote the number of positive primes less than or equal x . Thus $\pi(1) = 0$, $\pi(3.5) = 2$, and $\pi(11) = 5$. Now I am going to give you $\pi(x)$ for an increasing sequence of x 's and you are to compute $q(x) = x/\ln x$. Then you are to compare $\pi(x)$ with $q(x)$ by looking at the quotient, $Q(x) = \pi(x)/q(x) = (\pi(x) \ln x)/x$. Then you are to make a conjecture about $\lim_{x \rightarrow \infty} Q(x)$. (Note: $\ln x$ denotes the natural logarithm of x .) Here is the sequence:

$$\begin{array}{ll} \pi(1000) = 168, & \pi(10,000) = 1229, \\ \pi(100,000) = 9,592, & \pi(1,000,000) = 78,498, \\ \pi(10,000,000) = 664,579, & \pi(100,000,000) = 5,761,455. \end{array}$$

Now what do you think $Q(x)$ does as x gets big? You have conjectured the **Prime Number Theorem**, the proof of which is among the greatest of all mathematical achievements. It is beyond us at this point. The theorem was proved independently by J. Hadamard and C.J. de la Vallée-Poussin in 1896, using important results in complex analysis developed by G.F.B. Riemann, in particular the Riemann Zeta function, which we will define later.

2.2 Some Special Primes in $\mathbb{Z}$

Exercise 2.17. Let n be a positive integer. If $2^n + 1$ is prime, then n is a power of 2. (Hint: $\frac{1+2^{12}}{1+2^4} = 1 - 2^4 + 2^8$; you can generalize.)

Definition. Let n be a nonnegative integer and let $F_n = 2^{2^n} + 1$. Then F_n is said to be a **Fermat number** (a **Fermat prime** if F_n is prime).

Exercise 2.18. F_0, F_1, F_2 , and F_3 are all primes.

Exercise 2.19. Fermat thought that F_n is prime for all n . Show, as did Euler, that 641 is a factor of F_5 . You can do it! In fact, F_0, F_1, F_2, F_3 , and F_4 are the only Fermat numbers known to be prime! There may or may not be others.

We will have a bit more to say about Fermat primes when we study the Euler ϕ -function. At this point I hope that you are going to be surprised when I tell you that they are intimately connected with the problem of the constructibility (with straight edge and compass) of regular polygons. Have you ever constructed a regular triangle? A square? A regular hexagon? Of course you have. How about a regular pentagon? This one is harder, but I am confident that you could find a way to do it. However, you have not constructed (with straight edge and compass, of course) a regular 7-gon or a regular 11-gon. How do I know? I will tell you more about it later. Now, having searched the set of positive integers of the form $2^n + 1$ for primes, we change the sign between the terms:

Definition. Let n be a positive integer. Then $M_n = 2^n - 1$ is a **Mersenne number**. If M_n is prime, then M_n is a **Mersenne prime**.

Exercise 2.20. If M_n is prime, then n is prime. Hint: $(2^{kr} - 1)/(2^r - 1) = 2^{(k-1)r} + 2^{(k-2)r} + \dots + 2^r + 1$.

Exercise 2.21. Find a few Mersenne primes.

As in the case of Fermat primes, we don't know whether the set of Mersenne primes is infinite; large primes are central to modern communication by secret code. Newly discovered enormously large Mersenne primes are announced frequently.

Mersenne primes are closely connected with "perfect" numbers:

Definition. Let n be a positive integer and let $\tau(n)$ denote the number of positive divisors of n while $\sigma(n)$ denotes the sum of these divisors. (Thus, for example, $\tau(9) = 3$ and $\sigma(9) = 13$.) A **perfect number** is a positive integer n such that $\sigma(n) = 2n$. (The sum of all divisors which are less than n is n .)

Exercise 2.22. Find the two least perfect numbers.

Exercise 2.23. Let p be a positive prime and r a positive integer. Then $\tau(p^r) = r + 1$ and $\sigma(p^r) = 1 + p + p^2 + \dots + p^r = \frac{p^{r+1} - 1}{p - 1}$.

Exercise 2.24. Let m and n be relatively prime positive integers, let D_m denote the set of all positive divisors of m , let D_n denote the set of all positive divisors of n , and let D_{mn} denote the set of all positive divisors of mn . Then

$$D_{mn} = \{ab : a \text{ is in } D_m \text{ and } b \text{ is in } D_n\}.$$

Definition. A function of f from the positive integers to the complex numbers is **multiplicative** if $f(mn) = f(m)f(n)$ for relatively prime positive integers m and n .

Exercise 2.25. Give examples of multiplicative functions and examples of functions from the positive integers to the complex numbers that are not multiplicative. (You realize, of course, that a function from the positive integers to the integers is a function from the positive integers to the complex numbers.)

Exercise 2.26. Both τ and σ are multiplicative. (Hint: Exercise 2.24)

Exercise 2.27. Let n be a positive integer expressed (uniquely) as a product of powers of positive primes. Use Exercise 2.23 and 2.26 to find formulas for $\tau(n)$ and $\sigma(n)$.

Exercise 2.28. (Euclid) Let M_p be a Mersenne prime. Then $2^{p-1}M_p$ is a perfect number.

Exercise 2.29. (Euler) Let n be an even perfect number. Then $n = 2^{p-1}M_p$ for some Mersenne prime, M_p . (Let us break this exercise down a bit):

A) Let n be even and perfect. Then $n = 2^r m$, where m is odd.

B) $(2^{r+1} - 1)\sigma(m) = 2^{r+1}m$, so that $2^{r+1} | \sigma(m)$ and $(2^{r+1} - 1) | m$.

C) $\sigma(m) = 2^{r+1}s$ and $m = (2^{r+1} - 1)t$ for some integers s and t .

D) $(2^{r+1} - 1)2^{r+1}s = 2^{r+1}(2^{r+1} - 1)t$, so that $s = t$.

E) $\sigma(m) = m + s$, where $s | m$. This implies something special about m . Proceed.

Exercise 2.30. Use Exercise 2.28 to help you find a third (even) perfect number.

Exercise 2.31. In the proof of Exercise 2.29, where was it necessary to use the hypothesis that n is even?

(No one knows whether there exist odd perfect numbers, and since we don't know whether there exist infinitely many Mersenne primes, we don't know whether there are infinitely many even perfect numbers.)

For an exhaustive discussion of the lore and history of perfect numbers, see L.E. Dickson's *History of the Theory of Numbers*, Vol. 1.

Chapter 3

Congruences

3.1 Congruences and the Ring $\mathbb{Z}_n$

Definition. Let J be an integral domain and let a , b , and m be in J and $m \neq 0$. The statement that a is **congruent to b (modulo m)** means that $a - b$ is divisible (in J) by m . This is written: $a \equiv b \pmod{m}$.

Congruences were invented by Gauss around 1800; they became indispensable in the study of numbers.

Exercise 3.1. In J , $a \equiv a \pmod{m}$; if $a \equiv b \pmod{m}$, then $b \equiv a \pmod{m}$; if $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$, then $a \equiv c \pmod{m}$.

Exercise 3.2. In J , if $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then $a + c \equiv b + d \pmod{m}$, $a - c \equiv b - d \pmod{m}$, and $ac \equiv bd \pmod{m}$. If $a = mq + r$, then $a \equiv r \pmod{m}$.

Exercise 3.3. In $\mathbb{Z}$, $a \equiv b \pmod{m}$ if and only if $a = mq + r$ and $b = mQ + r$, where $0 \leq r < |m|$. (That is, a and b leave the same remainder when divided by m .)

Exercise 3.4. In $\mathbb{Z}$ find all x between 0 and 6 inclusive such that $3x \equiv 1 \pmod{7}$.

Exercise 3.5. Let m be in $\mathbb{Z}$, $m \neq 0$, and let a be in $\mathbb{Z}$. Then a is congruent mod m to one and only one member of the set, $\{0, 1, 2, \dots, |m| - 1\}$ and this member can be found by division. Give examples.

Exercise 3.6. In G , $3 + 2i \equiv 1 \pmod{1 + i}$ and $5 + 10i \equiv 0 \pmod{1 + 2i}$.

Exercise 3.7. Let a be in G and let $\delta = 1 + i$. Then a is congruent mod δ to one and only one member of the set, $\{0, 1\}$, and a is congruent mod 2 to one and only one member of the set, $\{0, 1, i, 1 + i\}$. In each case this member can be found by division. Give examples.

Exercise 3.8. In $\mathbb{Q}[x]$, $x^2 + 4x + 5 \equiv 2 \pmod{x + 1}$.

Exercise 3.9. Let $f(x)$ be in $\mathbb{Q}[x]$. Then $f(x)$ is congruent mod $x^3 + 1$ to one and only one polynomial in $\mathbb{Q}[x]$ with degree less than 3 or with no degree, and this polynomial can be found by division. Give examples.

Exercise 3.10. In $\mathbb{Z}$ find an x that satisfies all four of the following conditions simultaneously: $x \equiv 1 \pmod{2}$, $x \equiv 2 \pmod{3}$, $x \equiv 3 \pmod{5}$, $x \equiv 4 \pmod{7}$.

Exercise 3.11. Let n denote a positive integer. Then $10^n \equiv 1 \pmod{9}$. (Use Ex. 3.2.) ⁷

Exercise 3.12. Let n denote a positive integer. Then n is congruent mod 9 to the sum of its digits. (Hint: What does 328 mean, for example? Does it mean $8 + 2(10) + 3(10^2)$? Make repeated use of Exercise 3.2.)

Exercise 3.13. What is the remainder when $(1327945386)(123456)$ is divided by 9? Don't do this the long way. After all, you can "cast out" nines: 2 and 7 sum to 0 mod 9, so do 4 and 5, etc. Do it fast!

Exercise 3.14. Making use of Exercise 3.2 and "casting out" nines, quickly check this multiplication for accuracy: $(38)(42) = 1696$.

Exercise 3.15. Let $n = 1 + 2 + 3 + \dots + 300$. Then $n \equiv 6 \pmod{9}$.

Exercise 3.16. The huge number gotten by writing down in order the numbers 1 through 300 is congruent mod 9 to 6.

Exercise 3.17. If $a \equiv 0 \pmod{9}$, then $a^2 \equiv 0 \pmod{9}$; if $a \equiv 1 \pmod{9}$, then $a^2 \equiv 1 \pmod{9}$, if $a \equiv 2 \pmod{9}$, then $a^2 \equiv 4 \pmod{9}$, ..., if $a \equiv 8 \pmod{9}$, then $a^2 \equiv 1 \pmod{9}$. You fill in all the missing pieces and conclude that if a is any integer, then a^2 is congruent mod 9 to one and only one of 0, 1, 4, or 7.

Exercise 3.18. If the positive integer n is not congruent mod 9 to one of 0, 1, 4, or 7, then n is not a square.

Exercise 3.19. The huge number of Exercise 3.16 is not a square. Neither is it a cube. Neither is it the sum of two squares.

Exercise 3.20. Let a be in $\mathbb{Z}$ and let $[a]_5$ denote the set of all integers that are congruent mod 5 to a . Then, for instance, $[3]_5 = \{\dots, -7, -2, 3, 8, 13, \dots\}$. Now, write down in a similar way: $[0]_5$, $[1]_5$, $[2]_5$, $[4]_5$, $[5]_5$, $[6]_5$, and $[17]_5$.

Now, consider the set, $\mathbb{Z}_5 = \{[a]_5 : a \text{ is in } \mathbb{Z}\}$. How many distinct members has this set?

Exercise 3.21. Prove:

1. a is in $[a]_5$,
2. If x is in $[b]_5$, and x is in $[a]_5$, then $[a]_5 = [b]_5$,

3. If x is in $\mathbb{Z}$, then x is in one and only one $[a]_5$, where $0 \leq a < 5$, and therefore $\mathbb{Z}_5 = \{[0]_5, \dots, [4]_5\}$.

Exercise 3.22. What would $\mathbb{Z}_{12}$ mean? $\mathbb{Z}_n$? How many members has it?

Exercise 3.23. Let n be a positive integer. Then $\mathbb{Z}_n = \{[a]_n : a = 0 \text{ or } 1 \text{ or } \dots \text{ or } n-1\}$.

Unless needed for clarification, the subscripts on the $[a]$'s are going to be omitted. Thus, when we are talking about members of $\mathbb{Z}_n$, $[a]_n$ is simply going to be denoted by $[a]$.

Exercise 3.24. In $\mathbb{Z}_n$, $[a] = [b]$ if and only if $a \equiv b \pmod{n}$. In the set $\mathbb{Z}_n$, we are going to define “addition” and “multiplication”:

$$[a] + [b] = [a + b] \text{ and } [a][b] = [ab].$$

We should check that we aren't being silly about this. Suppose, for example, that $n = 12$. Now, by our definition, $[2][3] = [6]$. Fine. But now, $[2] = [14]$ and $[3] = [-9]$. Do you see that $[a]$ can be represented by many different a 's, and so can b ? We have made our definitions of addition and multiplication in terms of representatives; maybe $[a] + [b]$ (or $[a][b]$) is dependent on the particular a and b that are used to represent these sets. This would be unpleasant.

Exercise 3.25. In $\mathbb{Z}_n$, if $[a] = [a']$ and $[b] = [b']$, then $[a + b] = [a' + b']$ and $[ab] = [a'b']$; our definitions are not representative dependent.

Exercise 3.26. $\mathbb{Z}_n$ is an abelian group relative to addition.

Exercise 3.27. (i) Multiplication in $\mathbb{Z}_n$ is associative. (ii) There is a multiplicative identity in $\mathbb{Z}_n$.

Exercise 3.28. In $\mathbb{Z}_n$, multiplication distributes over addition.

Exercise 3.29. $\mathbb{Z}_n$ is a commutative ring with unity, the **ring of integers (mod n)**.

Exercise 3.30. Make operation (addition and multiplication) tables for $\mathbb{Z}_5$ and for $\mathbb{Z}_6$.

Exercise 3.31. $\mathbb{Z}_5$ is a field. $\mathbb{Z}_6$ is not an integral domain and is, therefore, not a field.

Exercise 3.32. If F is a field, the nonzero members of F are a multiplicative group.

Exercise 3.33. Let U_6 denote those members of $\mathbb{Z}_6$ that have multiplicative inverses. Then U_6 is a multiplicative group.

Exercise 3.34. Generalize Exercise 3.33: Let n be an integer greater than 1 and let U_n denote the subset of $\mathbb{Z}_n$ each of whose members has a multiplicative inverse. Then U_n is a multiplicative group, and $U_n = \{[a] : 1 \leq a \leq n-1$

and $\gcd(a, n) = 1$. (Hint: If $\gcd(a, n) = 1$, then $ax + ny = 1$, then $ax \equiv 1 \pmod{n}$, then $[ax] = [1]$, then $[a][x] = [1]$. This is not the whole proof!)

Exercise 3.35. Write down the members of $\mathbb{Z}_{12}$ and those of U_{12} . Make a multiplication table for U_{12} .

Exercise 3.36. If p is prime, then U_p consists of all nonzero members of $\mathbb{Z}_p$ and therefore $\mathbb{Z}_p$ is a field. If n is not prime, then $\mathbb{Z}_n$ has some nonzero members that are not in U_n and therefore $\mathbb{Z}_n$ is not a field.

3.2 The Euler ϕ -Function

Definition. The **order of a group** is the number of members in the group. Let n be an integer greater than 1. We let $\phi(n)$ denote the order of the group, U_n . If $n = 1$, then we define $\phi(n) = 1$. Here, ϕ is called the **Euler ϕ -function**.

Exercise 3.37. If n is a positive integer, then $\phi(n)$ is the number of positive integers less than or equal to n and relatively prime to n .

Exercise 3.38. Find $\phi(n)$ for $n = 1$, for $n = 2, \dots$, for $n = 24$.

The Euler ϕ -function is important in the study of numbers. We shall have to find a formula giving $\phi(n)$ for any positive integer n . (Recall that you did this for σ and τ .) You can do it right now for n a power of a prime:

Exercise 3.39. Let p be a positive prime and n a positive integer. Then, counting the number of positive integers less than p^n and relatively prime to p , we find

$$\phi(p^n) = p^n - p^{n-1} = p^{n-1}(p - 1).$$

Now, wouldn't it be pleasant if ϕ were multiplicative? Why?

Exercise 3.40. Assuming that ϕ is multiplicative, find $\phi(48)$ and $\phi(1000)$.

We shall have to do a bit of work to find that ϕ is multiplicative.

Exercise 3.41. Let G be a group and H a nonempty subset. If H is closed relative to the operation (for x and y in H , xy (this means x operate y) is in H), and H has the property that when x is in H , the inverse of x is also in H , then H is a subgroup of G .

Exercise 3.42. Find a nontrivial subgroup of $\mathbb{Z}$ (addition, of course). Find one of $\mathbb{Z}_6$ (addition, of course). Find one of $\mathbb{Z}_5$ (addition). See how many subgroups of U_5 that you can find. What are the orders?

Definition. A group is **cyclic** if there exists in the group a member g such that every member of the group is a "power" of g . (Here one has to be a bit

careful about what is meant by a power of g . For example, if a is a member of G , then a^2 means a operate a . If the operation happens to be addition, then this means $a + a$; a^0 means the identity member of the group, and a^{-1} means the inverse of a , while a^{-2} means $(a^{-1})^2$.) The member g is called a **generator** of the group.

Exercise 3.43. The additive group of $\mathbb{Z}_n$ is cyclic, $[1]$ being a generator.

Exercise 3.44. The multiplicative group, U_5 , of $\mathbb{Z}_5$ is cyclic, $[2]$ being a generator.

Exercise 3.45. U_8 is not cyclic. $\mathbb{Z}$ is cyclic. U_9 and U_{18} are cyclic. U_{12} is not cyclic.

Exercise 3.46. Let G be any group (not necessarily cyclic and not necessarily finite). Let a be in G . Let $\langle a \rangle$ denote the set of all powers (positive, negative, and zero) of a . Then $\langle a \rangle$ is a cyclic subgroup of G (called the **subgroup generated by a**).

Definition. Let a be a member of the group G . The **order** of a is the order of the subgroup of G generated by a .

Exercise 3.47. In $\mathbb{Z}$ find (2) and (6) . In U_{13} find $([2])$ and $([3])$. Find the orders of all these members.

Exercise 3.48. Let G be finite and let a be in G . For some positive integer k , $a^k = e$, where e is the identity member of G . Hence, there is a least positive integer m such that $a^m = e$. Hint: If n is the order of G , then in the set, $\{e, a, a^2, a^3, \dots, a^n\}$, there is a repetition.

Exercise 3.49. Let G, a , and m be as in Ex. 3.48. The subgroup of G generated by a is $\{e, a, a^2, a^3, \dots, a^{m-1}\}$. The order of a is m , the least positive integer such that $a^m = e$.

Exercise 3.50. Go back to U_{13} and find the order of each member. Do the same for the members of U_8 . You should note that in each case in which the group is of finite order, the order of a member (and of the subgroup generated by that member) is a divisor of the order of the group.

We are going to prove that the order of a subgroup of a finite group divides the order of the group. This is one of the most basic and important facts about finite groups.

Definition. Let G be a group (not necessarily finite) and H a subgroup. For any a in G , let $aH = \{ah : h \text{ is in } H\}$, be called a **left coset of H in G** .

Exercise 3.51. Let G be $\mathbb{Z}$ and let $H = (5)$. Find $0H, 1H, 2H, 3H, 4H$, and $5H$. (Remember now that the operation in $\mathbb{Z}$ is addition so that aH really means $a + H$.) Have you seen this before? Is $a \equiv b \pmod{m}$ if and only if a and b are in the same coset of (m) in $\mathbb{Z}$?

Exercise 3.52. You are going to think that you are doing Ex.'s 3.21 and 3.23 again. Let G be a group and H a subgroup. Prove

1. If a is in G , then a is in aH .
2. If x is in aH and x is in bH , then $aH = bH$.
3. If G is finite (so, then, is H), and the order of H is m , then each aH contains exactly m members.
4. If G has order n and H has order m and there are k distinct aH 's, then $mk = n$, and thus $m|n$. (**Lagrange's Theorem**)

Exercise 3.53. Can a 12-member group have a subgroup of order 5? Can a group of order p (prime) have a subgroup of order other than 1 or p ?

Exercise 3.54. Let G be a group of order p (prime). Let a be in G and $a \neq e$. What is a 's order? A group of prime order is [fill in the blank]?

Exercise 3.55. Let G be a group of order n and let a be in G . Then $a^n = e$. (Hint: Let m be the order of a . What is a^m ? Does $m|n$? What is a^n ?)

Exercise 3.56. Let p be prime and let $[a]$ be in U_p . Then $[a]^{p-1} = [1]$. (Hint: What is the order of the group U_p ?)

Exercise 3.57. (Fermat's Little Theorem) Let p be a prime and p not divide a . Then $a^{p-1} \equiv 1 \pmod{p}$.

Exercise 3.58. Let $[a]$ be in U_n . Then $[a^{\phi(n)}] = [1]$.

Exercise 3.59. Let $\gcd(a, n) = 1$. Then $a^{\phi(n)} \equiv 1 \pmod{n}$. (**Euler's generalization**).

Exercise 3.60. What is the remainder when 150^{18} is divided by 19? What is the remainder when 159^{1000} is divided by 50?

(The Little Theorem can be used to show a positive integer composite: if n is a positive integer and one can find a positive integer, a , such that $a^{n-1} \not\equiv 1 \pmod{n}$, then n is not prime. You are probably aware that there are numbers known to be composite but whose factors are unknown. Do you see how this might be?)

The converse of the Little Theorem is false. Nevertheless there are some tests for primes based on the Little Theorem. To illustrate, we take a ridiculously simple case: $n = 29$. Suppose we don't know that 29 is prime. Let $a = 2$ (because powers of 2 are easy to manage). You can check that $2^{28} \equiv 1 \pmod{29}$. ($2^5 \equiv -3$, $2^{10} \equiv 9$, etc.) This does not prove that 29 is prime, but it does show that the order of $[2]$ in U_{29} is a divisor of 28. Now $2^{14} \not\equiv 1 \pmod{29}$ and $2^4 \not\equiv 1 \pmod{29}$. This is sufficient to prove that 28 is the order of $[2]$, for if 2^7 or 2^2 were congruent mod 29 to 1, then so would be

2^{14} or 2^4 , respectively. Thus, 29 is prime. Why? If 2^{14} had turned out to be congruent mod 29 to 1, would this have proved that 29 isn't prime?

Note: We are still working on the question of whether the Euler ϕ -function is multiplicative. There are faster routes, but not as instructive.

Exercise 3.61. Prove:

- 1) If m and n are relatively prime and c is any integer, then there exist x and y in $\mathbb{Z}$ such that $xm + yn = c$.
- 2) If m, n , and c are in $\mathbb{Z}$, then there exist x and y in $\mathbb{Z}$ such that $xm + yn = c$ if and only if $\gcd(m, n) \mid c$.

Exercise 3.62. 1) If m and n are relatively prime and a and b are in $\mathbb{Z}$, then there exists x in $\mathbb{Z}$ such that $x \equiv a \pmod{m}$ and $x \equiv b \pmod{n}$. Hint: Use Exercise 3.61 to show that $a + km = b + rn$ can be solved for r and k .

Exercise 3.63. Show that the x guaranteed by Exercise 3.62 is unique mod mn . That is, show that if x and y are two integers each of which is congruent mod m to a and congruent mod n to b , then $x \equiv y \pmod{mn}$.

Exercise 3.64. If m, n , and k are relatively prime in pairs and a, b , and c are in $\mathbb{Z}$, then there exists x in $\mathbb{Z}$ such that $x \equiv a \pmod{m}$, $x \equiv b \pmod{n}$, and $x \equiv c \pmod{k}$. Give examples. Then generalize. You will then have the **Chinese Remainder Theorem** in $\mathbb{Z}$. You can also prove a similar theorem in the Gaussian Integers and in polynomials over a field.

Exercise 3.65. Let $S = \{1, 2\}$ and $T = \{1, 2, 3\}$. For each s in S and t in T , find an x such that $x \equiv s \pmod{3}$ and $x \equiv t \pmod{4}$. Find what x is congruent mod 12.

Exercise 3.66. If $\gcd(x, m) = 1$ and $\gcd(x, n) = 1$, then $\gcd(x, mn) = 1$.

Definition. Let each of G and L be a group, and $\mathbf{G} \times \mathbf{L} = \{(a, b) : a \text{ is in } G \text{ and } b \text{ is in } L\}$. For (a, b) and (c, d) in $G \times L$ we define the "product" to be (ac, bd) , where these indicated products are taking place in G and L , respectively.

Exercise 3.67. Make an operation table for $U_3 \times U_4$.

Exercise 3.68. The operation defined above makes $G \times L$ into a group.

Definition. A mapping f from a group G to a group G' is a **homomorphism** if $f(ab) = f(a)f(b)$ for a and b in G . If f is also bijective, then f is an **isomorphism**, and G and G' are **isomorphic**. If f is a mapping from a field F to a field F' such that f is an isomorphism between the two additive groups and also between the two multiplicative groups (units) then f is a **field isomorphism**. Isomorphic groups (fields) are, from the point of view of their operation tables, just alike except for the symbols used to represent members.

Exercise 3.69. Let G be $\mathbb{Z}_4$ (addition) and let G' be the (multiplicative) fourth roots of 1. Find an isomorphism from G to G' . Is your answer unique? How many such isomorphisms do you think there are?

Exercise 3.70. The function, $f(x) = \ln(x)$ is a group isomorphism (between what groups?). The function g defined between exercises 1.100 and 1.101 is a field isomorphism from $\mathbb{C}$ to $\mathbb{C}$ (an **automorphism**).

Exercise 3.71. $U_3 \times U_4$ is isomorphic to U_{12} . Check that f is an isomorphism, where

$$\begin{aligned} f([1]_3, [1]_4) &= [1]_{12}, \\ f([1]_3, [3]_4) &= [7]_{12}, \\ f([2]_3, [1]_4) &= [5]_{12}, \\ f([2]_3, [3]_4) &= [11]_{12}. \end{aligned}$$

How do you think I got this? You should go back and look at Exercise 3.65.

Exercise 3.72. This is a dinky little result that you will need soon: If $x \equiv y \pmod{n}$ and $\gcd(x, n) = 1$, then $\gcd(y, n) = 1$.

Exercise 3.73. Here we are! Let m and n be relatively prime. Then $U_m \times U_n$ is isomorphic to U_{mn} and therefore $\phi(mn) = \phi(m)\phi(n)$.

Exercise 3.74. Find $\phi(n)$ for $n = 468$, for $n = 265$, for $n = 1,000,000,000$. For what positive integers n is $\phi(n)$ odd?

Exercise 3.75. If $n = 2^k p_1 p_2 \dots p_r$, where each p is a Fermat prime and no two of the p 's are equal, then $\phi(n)$ is a power of 2. Conversely, if $\phi(n)$ is a power of 2, then n has the given form. (A regular polygon with n sides is constructible with straight edge and compass if and only if $\phi(n)$ is a power of 2. This striking result, due to Gauss, is proved by relating the geometric process of locating points in the plane to the algebraic one of extending the rational field of Q to certain larger subfields of the complex numbers. We won't prove it, although it is not beyond our capability.)

Exercise 3.76. Give some n 's for which regular polygons with n sides are constructible, and give some for which they are not.

Exercise 3.77. Check that $\sum_{d|6} \phi(d) = 6$, $\sum_{d|15} \phi(d) = 15$, and $\sum_{d|24} \phi(d) = 24$.

Make a conjecture.

Exercise 3.78. Let i denote the function from the positive integers to the complex numbers (to $\mathbb{Z}$, to be more exact) such that $i(n) = \sum_{d|n} \phi(d)$. If p is a positive prime, and k is a nonnegative integer, then $i(p^k) = p^k$.

Exercise 3.79. If f is a multiplicative function and $F(n) = \sum_{d|n} f(d)$, then F is multiplicative. This implies that the function i of Exercise 3.78 is multiplicative and the conjecture of Exercise 3.77 checks out. (You will find Exercise 2.24 helpful here.)

Exercise 3.80. There is a curious (at first glance) kinship among ϕ , τ , and σ :

$$\text{Let } \alpha(n) = \sum_{d|n} \phi(d) \tau(n/d).$$

Find $\alpha(6)$ and compare with $\sigma(6)$. Find $\alpha(18)$ and compare with $\sigma(18)$. Make a conjecture. The function α is not particularly important for our purposes. I think you could prove your conjecture but it might be hard. It happens that the proof will be an example of the fruit that can be picked easily after we have made a study of another integral domain, the domain of all functions from $\mathbb{Z}^+$ to the complex numbers. Of course we will have to define operations on the members of the set in order to have hope of making it into an integral domain.

3.3 Arithmetic Functions

A function from $\mathbb{Z}^+$ (the positive integers) to the complex numbers is called an **arithmetic function**. We let A denote the set of all such functions. We have seen four members, (τ , σ , ϕ , and i) of A that are multiplicative. (Of course the function, i , turned out to be a very simple function: $i(n) = n$.) We are now going to do a few exercises to show you that we can define addition and multiplication in A so as to make A an integral domain. We will then exploit the algebraic structure of A to obtain a classic and important result in number theory, the Möbius Inversion Formula.

Definition. Let f and g be in A . Then their sum, $f + g$, is that member of A such that

$$(f + g)(n) = f(n) + g(n),$$

and their **(Dirichlet) product**, $f * g$, is that member of A such that

$$(f * g)(n) = \sum_{d|n} f(d)g\left(\frac{n}{d}\right).$$

(The Dirichlet product of two members of A is far more useful than the sum. We define the sum largely because it is pleasing to have an integral domain.) We let $\mathbf{u}$, $\mathbf{z}$, and $\mathbf{e}$ denote members of A such that for all n , $u(n) = 1$ and $z(n) = 0$, while $e(n) = 0$ if $n \neq 1$ and $e(1) = 1$.

We can get a triple of easy pickings from the definitions and Exercise 3.79:

Exercise 3.81. Show that $u * \phi = i$, that $u^2 = \tau$, and that $u * i = \sigma$.

Now let us see whether our definitions of addition and multiplication in A are fruitful.

Exercise 3.82. A is an abelian group relative to addition.

Exercise 3.83. Multiplication in A is associative and commutative. Hint: $f * (g * h)(n)$ is the sum of all $f(a)g(b)h(c)$, where $abc = n$.

Now we can get the conjecture of Exercise 3.80 for a nickel:

Exercise 3.84. $\phi * \tau = \sigma$. Hint: $u * i = \sigma = u * (u * \phi)$.

Exercise 3.85. The member e of A is the unity (multiplicative identity).

Exercise 3.86. Multiplication in A distributes over addition.

Exercise 3.87. The member z of A is the zero member (additive identity) and if f and g are in A and $f * g = z$, then $f = z$ or $g = z$. (Hint: If $f \neq z$, there is a least n such that $f(n) \neq 0$.)

Exercise 3.88. A is an integral domain.

Exercise 3.89. The member f of A is a unit if and only if $f(1) \neq 0$.

Exercise 3.90. The functions, u , i , e , τ , σ , and ϕ , are units in A .

Exercise 3.91. If $f \neq z$ is in A and f is multiplicative, then f is a unit in A and f 's inverse is itself multiplicative. (Let g denote f 's inverse. If g is not multiplicative, there exists some least positive integer mn such that $\gcd(m, n) = 1$ and $g(m)g(n) \neq g(mn)$. Now use that $f * g = e$ and remember $g(ab) = g(a)g(b)$ if $\gcd(a, b) = 1$ and $ab < mn$. Write out the product of $(f * g)(m)$ with $(f * g)(n)$, using multiplicativity where possible, and compare with $(f * g)(mn)$. You might try $mn = 3 \times 4$.)

Exercise 3.92. The subset M of all nonzero multiplicative functions in A is a subgroup of the group of units of A .

Exercise 3.93. Find $\sigma^{-1}(p^k)$, where p is prime. Find all n such that $\sigma^{-1}(n) = 2n$. Shall we call them (it) inverse perfect?

Exercise 3.94. Let μ denote the multiplicative inverse of the function u . The function μ is the **Möbius** function. Show that $\mu(1) = 1$, $\mu(2) = -1 = \mu(3)$, and $\mu(4) = 0$.

Exercise 3.95. Let p denote a prime in $\mathbb{Z}^+$. Then $\mu(p) = -1$, and $\mu(p^k) = 0$ if k is an integer greater than 1. (Use that $u * \mu = e$.)

Exercise 3.96. Use Exercise 3.92 to show that μ is in M and then use Exercise 3.95 to find a way to determine $\mu(n)$ for any positive integer n and thus get Möbius's definition of μ .

Exercise 3.97. If f and g are in A and $f = u * g$, then $g = \mu * f = f * \mu$.

Exercise 3.98. (Möbius Inversion Formula) If f and g are in A and

$$f(n) = \sum_{d|n} g(d), \text{ then } g(n) = \sum_{d|n} f(d) \mu\left(\frac{n}{d}\right).$$

Show that if f is multiplicative, then so is g .

Exercise 3.99. By Exercise 3.79, $i = \phi * u$ and Exercise 3.97 implies that $\phi = \mu * i$.

We will make use of this inversion formula later in our study. At this point I think I should show you an example of its use in the inversion (in the usual sense) of functions whose domains are not the positive integers, but rather the real or complex numbers.

The **Riemann zeta function** is defined this way:

$$\zeta(z) = \sum_{n=1}^{\infty} \frac{1}{n^z},$$

for complex numbers z with real part $x > 1$.

(I am using z as the variable here; it is customary to use s instead.) The zeta function is an important one; among other uses it is central to the 1896 proofs of the prime number theorem which I told you about. Now,

$$\frac{1}{\zeta(z)} = \sum_{n=1}^{\infty} \frac{\mu(n)}{n^z},$$

the likely truth of which you can probably convince yourself by writing out a few terms of $\zeta(z)$ and of the other series and multiplying.

It is also true that $\sum_1^{\infty} \frac{\mu(n)}{n} = 0$. You would probably have a hard time convincing yourself of this because it is equivalent to the prime number theorem in the sense that each implies the other!

Exercise 3.100. Use your result of Exercise 3.95 as the definition of μ and prove the inversion formula independently of the algebraic machinery above.

3.4 Primitive Roots (mod p)

Exercise 3.101. U_n is cyclic for $n = 2, n = 3, n = 4, n = 5, n = 6$, and $n = 7$.

Exercise 3.102. U_8 is not cyclic; neither is U_{20} .

Definition. If U_n is cyclic and $[a]$ is a generator, then a is said to be a **primitive root (mod n)**.

We are going to do some exercises to show that if p is prime, then U_p is cyclic; in other words that there exists a primitive root mod p . In fact, we shall show that U_p has $\phi(p-1)$ primitive roots if p is prime.

Exercise 3.103. Find a primitive root mod p for each of these primes: 3, 5, 7, 11, 13, 17.

Exercise 3.104. Let G be a group and let a in G have order m . If k is a positive integer and $a^k = e$, then $m|k$. (Hint: $k = qm + r$.)

Before proving our theorem, we do a couple of exercises that shed some light on the structure of cyclic groups.

Exercise 3.105. If G is a cyclic group of order n and g is a generator of G (and thus $G = \{e, g, g^2, \dots, g^{n-1}\}$), then the order of g^k is $n/\gcd(k, n)$. It follows that G has $\phi(n)$ generators. (Hint: Let $n_1 = n/\gcd(k, n)$. First show that $(g^k)^{n_1} = e$, so that the order, t , of g^k divides n_1 . Next, since $(g^k)^t = e$, n divides kt . Use this to get that n_1 divides t .)

Exercise 3.106. If G is cyclic of order n and $d|n$, then G has only one subgroup of order d . (Hint: Let $dk = n$ and let g be a generator of G . Verify that g^k has order d and therefore generates a subgroup of order d . Then show that any member of G having order d is in this subgroup.) Show that any subgroup of a cyclic group is cyclic.

Exercise 3.107. Let F be a field (we will specialize to $\mathbb{Z}_p$ shortly). Remind yourself of Exercise 1.97: If a is in F and $f(x)$ is in $F[x]$, then $f(a) = 0$ iff $(x - a) | f(x)$ in $F[x]$.

Exercise 3.108. If f of Exercise 3.107 has degree n , there are at most n members a of F such that $f(a) = 0$.

Exercise 3.109. Every nonzero member of $\mathbb{Z}_p$ satisfies $x^{p-1} - [1] = [0]$, and this polynomial factors into $p-1$ distinct linear factors in $\mathbb{Z}_p[x]$. If d is a positive divisor of $p-1$, then $x^d - [1]$ also factors into d distinct linear factors; thus d distinct members of $\mathbb{Z}_p$ are zeros of $x^d - [1]$.

Exercise 3.110. Let d be a positive divisor of $p-1$, and let C denote the set of all positive divisors of d , while R denotes the set of all zeros of $x^d - [1]$ in $\mathbb{Z}_p$. The order (in U_p) of each member of R is a unique member of C .

Exercise 3.111. Let $p = 13$ and $d = 4$. List the set R and set C . Draw arrows from R to C , connecting each member, r , of R , with a member of C (the order of r as a member of U_{13}).

Exercise 3.112. Let c be a member of C and let $\psi(c)$ be the number of members of R that have order c . Then if we let $f(d) = \sum_{c|d} \psi(c)$, we have that $f(d) = d$, since R has d members. Check this in your example.

Exercise 3.113. $\psi(d) = \sum_{c|d} \mu(c) f(d/c) = \sum_{c|d} \mu(c)(d/c) = (\mu * i)(d) = \phi(d)$, by the Möbius inversion formula and Exercise 3.99.

Exercise 3.114. Since $p - 1$ is a divisor of $p - 1$, $\psi(p - 1) = \phi(p - 1)$, so U_p has members with order $p - 1$ and is therefore cyclic and there exist primitive roots mod p .

Exercise 3.115. Make a conjecture about which U_n 's are cyclic. That is for which n 's is there a primitive root mod n ?

We could do a sequence of exercises to answer this question, but semesters are cyclic and finite and we have other things to do in this cycle. It happens that U_n is cyclic iff n is a power of an odd prime or if $n = 2$ or 4 or twice a power of an odd prime.

The ϕ -function and primitive roots mod n are germane to the study of repeating decimal fractions. We will do just enough to arouse your curiosity. Consider the decimal fraction $0.a_1a_2a_3\ldots$ for $1/n$, where $n > 1$. By the divisor theorem we get

$$10(1) = a_1n + r_1, \quad r_1 < n, \quad r_1 \equiv 10 \pmod{n},$$

$$10r_1 = a_2n + r_2, \quad r_2 < n, \quad r_2 \equiv 10^2 \pmod{n},$$

etc.

We see that $r_k \equiv 10^k \pmod{n}$.

Exercise 3.116. Find the decimal fraction for each $k/7$, where $0 < k < 7$.

Exercise 3.117. Suppose that $\gcd(10, n) = 1$. Then the number of distinct remainders in the above sequence (preceding Exercise 3.116) is the order of $[10]$ in U_n . The length of the repetend is a divisor of $\phi(n)$, and the length is $\phi(n)$ iff $[10]$ is a generator of U_n .

Exercise 3.118. Under what circumstances will the decimal for k/n be a permutation of the digits of that for $1/n$?

We will end this discussion here, although you can see that there are many questions whose answers we could chase. Name some.

3.5 Communicating by Secret Code

I told you earlier that huge primes were central to modern communication by secret code. Now we have built the machinery (congruences and Euler's generalization of Fermat's Little Theorem) to see how this works. I will give you the theory only, with no attempt to discuss the computational aspects.

Suppose you and I are in a network, the members of which communicate by secret code, and that we have adequate computer power (it exists) to carry out the operations that I am going to describe. You select two very large primes, p and q , whose product is n , a number so big that even the most powerful computers are unable to factor it in a tolerable length of time. Only you know the factors of n , but you publish n itself in a directory. You also select and publish a positive integer k that is coprime with $\phi(n)$. Since $\gcd(\phi(n), k) = 1$, there is a positive integer m such that $km \equiv 1 \pmod{\phi(n)}$; put another way, $[k][m] = [1]$ in $U_{\phi(n)}$. You publish both n and k , but you keep p , q , and m to yourself.

Now, let us say that I want to send you a secret message. I first convert the message to a number, M , by means of a numerical alphabet. For example, here is standard one:

$A \rightarrow 01,$	$, (\text{comma}) \rightarrow 27,$
$B \rightarrow 02,$	$. (\text{period}) \rightarrow 28,$
$C \rightarrow 03,$	$? \rightarrow 29,$
$\dots,$	$0 \rightarrow 30,$
$\dots,$	$1 \rightarrow 31,$
$I \rightarrow 09,$	$\dots,$
$J \rightarrow 10,$	$\dots,$
$K \rightarrow 11,$	$9 \rightarrow 39,$
$\dots,$	$(\text{with } 00 \text{ indicating space between words})$
$\dots,$	
$Z \rightarrow 26.$	

For example ⁸ *I passed my comps.* would be numberized this way:

$$M = 090016011919050400132500031513161928$$

It is assumed that the numerical message, M , is a number less than n . (Messages can be broken into blocks if desirable.) We also assume that $\gcd(M, n) = 1$. To send you the message, M , I look up your n and k , find M^k , and reduce mod n . That is, $M^k \equiv r \pmod{n}$. I then send you the number r . You then find r^m .

Exercise 3.119. $r^m \equiv M^{km} \pmod{n}$

Exercise 3.120. $r^m \equiv M^{\phi(n)t+1} \equiv (M^{\phi(n)t})M \pmod{n}$, for some positive integer t , since $km \equiv 1 \pmod{\phi(n)}$.

Exercise 3.121. $M^{\phi(n)t} \equiv 1 \pmod{n}$. Then $r^m \equiv M \pmod{n}$. You have recovered my numerical message M , which you convert to English with the standard alphabet.

(The assumption that $\gcd(M, n) = 1$ was used in this proof. A proof can be made without this assumption if we retain the hypothesis that $M < n$ and note that at most one of p and q can divide n .)

Chapter 4

Quadratic Reciprocity

4.1 Squares mod p

We let S_p denote the squares in U_p , where p is prime in $\mathbb{Z}$. That is, $[a]$ is in S_p if and only if $[a] = [b]^2$ for some $[b]$ in U_p .

Exercise 4.1. Find S_7 and S_{17} .

Exercise 4.2. Find a generator of U_{17} and denote it by g . Now, find the set of all even powers of g . How does this set compare with S_{17} ?

Exercise 4.3. If g is a generator of U_p , then S_p is the set of all even powers of g . (Do you remember how we can be sure that U_p has a generator if p is prime?)

Exercise 4.4. If $p > 2$, S_p is a subgroup of U_p , and has $(p-1)/2$ members.

Definition. Let p be prime and let a be in $\mathbb{Z}$ and $\gcd(a, p) = 1$. We say that a is a **quadratic residue** mod p if there is some x in $\mathbb{Z}$ such that $x^2 \equiv a \pmod{p}$. If there exists no such x then we say that a is a quadratic non-residue.

Exercise 4.5. Let a be in $\mathbb{Z}$. Then a is a quadratic residue mod p if and only if $[a]$ is in S_p .

Exercise 4.6. Prove:

- A) Let $b \equiv a \pmod{p}$. If a is a quadratic residue mod p , then so is b .
- B) If both a and b are quadratic residues mod p , then so is ab .
- C) If both a and b are quadratic nonresidues mod p , then ab is a quadratic residue mod p .
- D) 1 is a quadratic residue mod p and if $\gcd(a, p) = 1$, then a^2 is a quadratic residue mod p .

Definition. Let a be in $\mathbb{Z}$ and p be prime. Then (**Legendre's Symbol**): $\left(\frac{a}{p}\right) = 1$ if a is a quadratic residue mod p , -1 if a is a quadratic non-residue mod p , and 0 if $p|a$.

Exercise 4.7. Prove

- A) If $a \equiv b \pmod{p}$, then $\left(\frac{a}{p}\right) = \left(\frac{b}{p}\right)$,
- B) $\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right)$
- C) $\left(\frac{a^2}{p}\right) = 1$ if $\gcd(a, p) = 1$,
- D) $\left(\frac{1}{p}\right) = 1$.

Exercise 4.8. Find $\left(\frac{4}{7}\right)$, $\left(\frac{5}{7}\right)$, $\left(\frac{7}{5}\right)$, $\left(\frac{71}{5}\right)$, $\left(\frac{23}{5}\right)$, $\left(\frac{5}{23}\right)$, $\left(\frac{7}{11}\right)$, and $\left(\frac{11}{7}\right)$.

One can see that Exercise 4.7 implies that $\left(\frac{a}{p}\right)$ can be determined for any a provided $\left(\frac{q}{p}\right)$ can be determined for primes q . For odd primes $q \neq p$ this determination is given by the **Quadratic Reciprocity Law (QRL)**: For odd primes p and q , $\left(\frac{p}{q}\right) = \left(\frac{q}{p}\right)$ if at least one of p and q is congruent mod 4 to 1. If both are congruent mod 4 to -1 , then $\left(\frac{p}{q}\right) = -\left(\frac{q}{p}\right)$. To give you a sample of the efficiency of the law, let us do an exercise assuming the law and also assuming that $\left(\frac{2}{p}\right) = 1$ iff p is congruent mod 8 to ± 1 . (You will also need to use the results of Exercise 4.7).

Exercise 4.9. Find $\left(\frac{5}{23}\right)$, $\left(\frac{30}{89}\right)$, $\left(\frac{507}{773}\right)$, and $\left(\frac{27(507)}{773}\right)$. Is there an integer x such that $x^2 \equiv 507 \pmod{773}$? Note: 773 is prime, as you can determine by trying 3, 5, 7, 11, 13, 17, 19, 23, and 29 for divisors. (Why is this enough?) Similarly, you can search 507 for divisors. ⁹

Now we begin a sequence of exercises to prove the **QRL**. We are indebted to Gauss, who was first to find a proof. We will also determine $\left(\frac{-1}{p}\right)$ and $\left(\frac{2}{p}\right)$.

Exercise 4.10. Let p denote an odd prime and let a be an integer not divisible by p . Then $\left(\frac{a}{p}\right) \equiv a^{(p-1)/2} \pmod{p}$. Hint: $[a]^{(p-1)/2}$ is a sol'n in $\mathbb{Z}_p[x]$ of $x^2 - [1] = [0]$. There are only two sol'ns: $[1]$ and $[-1]$. Now let a be an even (odd) power of a generator of U_p . Proceed.

Exercise 4.11. Factor $x^2 - [1]$ completely into linear factors in $\mathbb{Z}_3[x]$. Factor $x^4 - [1]$ into linear factors in $\mathbb{Z}_5[x]$. Factor $x^{p-1} - [1]$ into linear factors in $\mathbb{Z}_p[x]$. Hint: See Ex. 3.109.

Exercise 4.12. Wilson's Theorem: If p is prime, then $(p-1)! \equiv -1 \pmod{p}$. Hint: put $x = [0]$ in the factorization above.

If $n > 1$ and $(n-1)! \equiv -1 \pmod{n}$, does it follow that n is prime?

Exercise 4.13. Gauss gave an elegant different proof of Wilson's theorem. Let us see an example and you can generalize. Let $p = 13$. Then $(p-1)! = 12! = (2 \times 7)(3 \times 9)(4 \times 10)(5 \times 8)(6 \times 11)(1 \times 12)$. Now reduce this product mod 13 and see what you get. I have grouped by what we now call inverses. Gauss called them associates. He did this theorem in his masterpiece, *Disquisitiones Arithmeticae*. He says that neither Wilson nor Waring, who attributed the theorem to Wilson, had a proof, that Waring implied that he was unable to effect a proof because *no notation could be devised to express a prime*. Then he says that truths of this kind (the theorem) should be drawn from notions rather than from notations. You must remember that we have handy algebraic tools that Waring and Wilson (and Gauss) lacked.

The truth of the **QRL** is surely not transparent, although almost anyone could conjecture it by examining a few odd primes. Had the proof not been difficult, others would have found one before Gauss, who worked hard on it. I am going to break it into modest steps so as to render the steps transparent (I hope). Again I resort to an example. Suppose we want to determine $\left(\frac{8}{13}\right)$, where we let $p = 13$ and $a = 8$. Look at this display of $(p-1)/2$ lines:

$$\begin{aligned} 1 \times 8 &= 0 \times 13 + 8 \\ 2 \times 8 &= 1 \times 13 + 3 \\ 3 \times 8 &= 1 \times 13 + 11 \\ 4 \times 8 &= 2 \times 13 + 6 \\ 5 \times 8 &= 3 \times 13 + 1 \\ 6 \times 8 &= 3 \times 13 + 9 \end{aligned}$$

Exercise 4.14. Find the product of the members of the left column, the product of the members of the right column and equate, then reduce mod p .

Ans.

$$\begin{aligned} \frac{p-1}{2}! a^{(p-1)/2} &\equiv 1 \times 3 \times 6 \times 8 \times 9 \times 11 \\ &\equiv 1 \times 3 \times 6 \times (13-5) \times (13-4) \times (13-2) \\ &\equiv \frac{p-1}{2}! (-1)^3 \pmod{p} \end{aligned}$$

Exercise 4.15. Cancel factorials (can you cancel across congruences?). Then

$$a^{(p-1)/2} \equiv (-1)^n \pmod{p},$$

where n is the number of remainders in the above display that exceed $p/2$.

Exercise 4.16. Continue with the above example. Exercise 4.15, with the help of Exercise 4.10, implies that $\left(\frac{8}{13}\right) = (-1)^3 = -1$. You should check this directly by listing the squares mod 13.

Now let's set out to prove the result suggested by the example which we just saw. We want to let p be an odd prime and a be an integer not divisible by p . We want to get the products, $1a, 2a, 3a, \dots, ((p-1)/2)a$ and write them as $ia = q_i p + r_i (i = 1, 2, 3, \dots, (p-1)/2)$. (Here, q_i denotes a quotient. Do not confuse it with a prime, q). I considered other notations for the quotient but couldn't (à la Waring) find a suitable one. Then we want to show that $\left(\frac{a}{p}\right) = (-1)^n$, where n is the number of the r_i 's that exceed $p/2$. It will then be necessary to find some way to manage $n \bmod 2$, but that's down the road a bit.

Exercise 4.17. Let p be an odd prime and p not divide the integer a . Consider the $(p-1)/2$ equations:

$$\begin{aligned} 1a &= q_1 p + r_1 \\ 2a &= q_2 p + r_2 \\ &\dots \\ &\dots \\ \left(\frac{p-1}{2}\right)a &= q_{(p-1)/2} p + r_{(p-1)/2}. \end{aligned} \quad 0 < r < p, \text{ for each } r.$$

A) The r 's are distinct.

B) For each r_i that exceeds $p/2$ let $r_i = p - s_i$ and let n denote the number of s_i 's. Then n s_i 's, together with the remaining r_i 's are the set, $\{1, 2, 3, \dots, (p-1)/2\}$ in some order.

C) Multiply by columns, cancel factorials, and get: $a^{(p-1)/2} \equiv (-1)^n \bmod p$.

D) $\left(\frac{a}{p}\right) = (-1)^n$. Note that this is equality, not just congruence.

The above exercise is known as **Gauss's Lemma**. You are probably saying, "What good is it? I don't know n ." As you will see, you don't have to know n ; of course we are going to use it somehow. We will do some clever counting mod 2 shortly, but before doing so we will use the lemma directly to determine $\left(\frac{-1}{p}\right)$ and $\left(\frac{2}{p}\right)$.

Exercise 4.18. Let p denote an odd prime. Then

$$\begin{aligned} 1(-1) &= (-1)p + (p-1) \\ 2(-1) &= (-1)p + (p-2) \\ &\text{etc.} \\ (p-1)/2(-1) &= (-1)p + (p+1)/2 \end{aligned}$$

All the r_i 's are greater than $p/2$ and $\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}$.

Exercise 4.19. If p is an odd prime, then $\left(\frac{-1}{p}\right) = 1$ iff $p \equiv 1 \pmod{4}$.

Don't be misled by the ease with which this result came. This is an important conclusion, which we used a big gun (Gauss's lemma) to get. Everyone interested in numbers commits it to memory. ¹⁰

Exercise 4.20. Take a few odd primes and check the result in Exercise 4.19. We can use the lemma directly to get $\left(\frac{2}{p}\right)$ for odd primes, p ; this is necessary if we hope to be able to determine $\left(\frac{a}{p}\right)$ for any a , because the **QRL** helps only if a is another odd prime.

Exercise 4.21. Now let $a = 2$ while p is an odd prime. Then,

$$\begin{aligned} 1 \times 2 &= 0p + 2 \\ 2 \times 2 &= 0p + 2 \times 2 \\ &\text{etc.} \\ i \times 2 &= 0p + 2i \\ &\text{etc.} \\ ((p-1)/2) \times 2 &= 0p + p-1. \end{aligned}$$

All the q 's are zero and $r_i = 2i$ for each i between 1 and $(p-1)/2$, inclusive.

Now we want n , the number of r 's exceeding $p/2$.

Exercise 4.22. Refer to Exercise 4.21. There, $r_i > p/2$ iff $i > p/4$. Thus n is the number of i 's between $p/4$ and $(p-1)/2$, including $(p-1)/2$.

Exercise 4.23. If p is an odd number then $p = 8k + R$, where $R = 1, 3, 5$, or 7 .

Exercise 4.24. Refer to 4.23. The least integer, i that is greater than $p/4$ is $2k+1$ if $R = 1$ or 3 and is $2k+2$ if $R = 5$ or 7 .

Exercise 4.25. If each of a and b is a positive integer and $b > a$, then the number of integers between a and b inclusive is $b - a + 1$.

Exercise 4.26. Refer to Exercise 4.22. Let $p = 8k + R$; the number n is
 $(4k) + (R - 1)/2 - (2k + 1) + 1 = 4k + (R - 1)/2 - 2k$ if $R = 1$ or 3
 and n is
 $(4k) + (R - 1)/2 - (2k + 2) + 1 = 4k + (R - 1)/2 - 2k - 1$ if $R = 5$ or 7 .

Exercise 4.27. Let R take its allowed values and conclude that

$$\left(\frac{2}{p}\right) = 1 \text{ iff } R = 1 \text{ or } 7 \text{ iff } p \equiv \pm 1 \pmod{8}.$$

We have gotten quite a bit of mileage from the display of Exercise 4.17 by multiplying by columns. We can get a great deal more if we add by columns. To move forward without a lot of wheel spinning, we need a definition. Let x denote a real number. Then the **greatest integer in** x is the greatest integer that does not exceed x . We will denote it by $\lfloor x \rfloor$, which is standard notation.

Exercise 4.28. Find $\lfloor 7.5 \rfloor$, $\lfloor \pi \rfloor$, $\lfloor -\pi \rfloor$, $\lfloor 29/3 \rfloor$.

Exercise 4.29. Let a, b, q , and r be in $\mathbb{Z}^+$ and let $a = bq + r$, with $0 \leq r < b$. Then $q = \lfloor a/b \rfloor$.

Exercise 4.30. Go back and rework the display of Exercise 4.17. Replace each q_i by $\lfloor (ia)/p \rfloor$. Let S denote $\sum_{i=1}^{(p-1)/2} i$. Add the left column to get Sa . Add the right column to get $p \times \left(\sum_{i=1}^{(p-1)/2} \lfloor (ia)/p \rfloor \right) + \sum r_i + np - \sum s_k$, where the r_i 's are those less than $p/2$ and the n s_k 's are gotten by subtracting the r_i 's that are greater than $p/2$ from p .

Exercise 4.31. Continue with Exercise 4.30. Let N denote the sum in the parentheses (multiplying p). Then

$$Sa = Np + np + \sum r_i - \sum s_k.$$

Exercise 4.32. Now assume a is odd (we know how to handle powers of 2) and remember that p is odd. Also note that -1 and 1 are congruent mod 2. Then since the r 's and s 's together make up the set $\{1, 2, 3, \dots, (p-1)/2\}$,

$$S \equiv N + n + S \pmod{2}.$$

Then $n \equiv N \pmod{2}$ and $\left(\frac{a}{p}\right) = (-1)^N$.

Exercise 4.33. Find $\left(\frac{3}{11}\right)$ by determining N directly from its definition. It may seem that we are no closer to our goal (the **QRL**), since we have merely transferred our attention from n to N , where n is the number of remainders

exceeding $p/2$ and N is the sum of the quotients. Let me point out that we are not trying to determine $\left(\frac{a}{p}\right)$ directly, but are trying to relate $\left(\frac{q}{p}\right)$ to $\left(\frac{p}{q}\right)$, where both are odd primes. We saw in Exercise 4.9 that if we could do so, then we could determine $\left(\frac{a}{p}\right)$.

Exercise 4.34. Check this restatement of Exercise 4.32, where a is now the odd prime, q .

If p and q are distinct odd primes and $N = \sum_{i=1}^{(p-1)/2} \lfloor (iq)/p \rfloor$, then $\left(\frac{q}{p}\right) = (-1)^N$.

Exercise 4.35. Let $M = \sum_{i=1}^{(q-1)/2} \lfloor (ip)/q \rfloor$. Then $\left(\frac{p}{q}\right) = (-1)^M$.

Exercise 4.36. In the notation of Exercises 4.34 and 4.35,

$$\left(\frac{q}{p}\right) \left(\frac{p}{q}\right) = (-1)^{N+M}.$$

Now you can see that we have to determine $N + M \pmod{2}$. We can do so with some simple geometry.

Exercise 4.37. Sketch the x, y plane and on it draw the lines, $x = p/2$ and $y = q/2$, and the diagonal line, $y = qx/p$.

Exercise 4.38. Let R denote the rectangle with boundaries $x = p/2, y = q/2, x = 0$, and $y = 0$. There are $(p-1)/2 \times (q-1)/2$ **lattice points** (points with integer coordinates) in the interior of the rectangle.

Exercise 4.39. There is no lattice point on the diagonal, $y = qx/p$.

Exercise 4.40. There are N lattice points in R below the diagonal, and there are M lattice points in R above the diagonal.

Exercise 4.41. $N + M = ((p-1)/2)((q-1)/2)$ and

$$\left(\frac{q}{p}\right) \left(\frac{p}{q}\right) = (-1)^{((p-1)/2)((q-1)/2)}.$$

Exercise 4.42. Now we have the long-awaited goal, the **QRL**: If p and q are odd primes, then

$$\left(\frac{q}{p}\right) = \left(\frac{p}{q}\right)$$

if and only if p or q is congruent mod 4 to 1.

Exercise 4.43. Find whether 244 is a quadratic residue mod 5, mod 3, mod 7, mod 11, mod 13, and mod 29.

Exercise 4.44. Exploit the **QRL** and a bit of ingenuity to determine whether there exists an integer x such that $3x^2 + 6x - 2 \equiv 0 \pmod{89}$.

Chapter 5

Sums of Two Squares and Pythagorean Triples

5.1 Which Positive Integers Are Sums of Two Squares?

Exercise 5.1. Check that the following is true: Every odd prime between 1 and 50 is the sum of two squares if and only if it is congruent mod 4 to 1. Make a conjecture.

Exercise 5.2. If the odd prime p is the sum of two squares, then $p \equiv 1 \pmod{4}$. (This is easy: Suppose $x^2 + y^2 = p$. Think about x^2 , y^2 , and $p \pmod{4}$.)

The Gaussian integers G provide a fruitful setting in which to study sums of two squares. Why? Well, a positive integer is a sum of two squares if it is the norm of a member of G . We propose to learn enough about G to be able to identify those norms.

Exercise 5.3. Prove:

- (A) 3, 7, 11, 19, 23, and 31 are all prime in G . (Consider norms.)
- (B) Neither of 5, 13, 17, and 29 is prime in G .
- (C) Each of the numbers in Exercise 5.3 (B) factors into a product of two distinct primes which are conjugates. By distinct, we mean that they are not associates. $5 = (1 + 2i)(1 - 2i)$, for example.
- (D) If p is prime in $\mathbb{Z}$ and $p \equiv 3 \pmod{4}$, then p is prime in G . (Suppose p factors in G . Then the product of the norms of the factors must be p^2 .)

Exercise 5.4. Prove:

- (A) Let p be prime in $\mathbb{Z}$ and $p \equiv 1 \pmod{4}$. Let $[t]$ be in $\mathbb{Z}_p$ such that $[t]^2 = [-1]$, where the brackets denote members of $\mathbb{Z}_p$. How do we know that there is such a t ? Then $t^2 \equiv -1 \pmod{p}$. Then $p \mid (t^2 + 1)$,

so that p divides the product $(t+i)(t-i)$. Does p divide either factor? Conclude that p is not prime in G .

(B) Let $p = 17$. Find a t such that $t^2 \equiv -1 \pmod{p}$. Verify that p divides $(t+i)(t-i)$ in G . Do the same for $p = 29$.

(C) Let p be prime in $\mathbb{Z}$ and $p \equiv 1 \pmod{4}$. Since p is not prime in G , let $p = \alpha\beta$ in G . Then $p^2 = N(\alpha)N(\beta)$ in $\mathbb{Z}$. It follows that $p = N(\alpha) = \alpha\bar{\alpha}$ in G , so that p is the sum of two squares in $\mathbb{Z}$. Both α and $\bar{\alpha}$ are prime in G and they are not associates.

Exercise 5.5. Factor 5, 13, 17, 29, 33, 37, and 41 into primes in G .

Exercise 5.6. In G ,

$$2 = (1+i)(1-i),$$

where both factors are primes. They are associates, so that, in fact, 2 is (except for a unit) the square of a prime in G .

Let us summarize what we know about primes in G :

- We know the prime factors of 2.
- We know that if p is prime in $\mathbb{Z}$ and $p \equiv 3 \pmod{4}$, then p is prime in G .
- We know that if p is prime in $\mathbb{Z}$ and congruent mod 4 to 1, then p factors into two distinct (and conjugate) primes in G .

Now we wonder whether there are other primes in G and hope there aren't.

Exercise 5.7. Let α be in G and $\alpha \neq 0$. Then $\alpha|N(\alpha)$ in G . $N(\alpha)$ is a positive member of $\mathbb{Z}$, so that there is a least positive integer divisible by α .

Exercise 5.8. If α is prime in G , then the least positive integer guaranteed by Exercise 5.7 is a prime in $\mathbb{Z}$. (Remember now: If $\alpha|\beta\chi$, then $\alpha|\beta$ or $\alpha|\chi$.)

Exercise 5.9. Let π be prime in G . Let L denote the (prime) least positive member of $\mathbb{Z}$ divisible by π .

- (A) If $L \equiv 3 \pmod{4}$, let L be denoted by q . Then $\pi = uq$, where u is a unit in G .
- (B) If $L \equiv 1 \pmod{4}$, let L be denoted by p . Then $p = \alpha\bar{\alpha}$, where both of these factors are prime and $\pi = u\alpha$ or $\pi = u\bar{\alpha}$, where u is a unit in G .
- (C) If $L \equiv 2 \pmod{4}$, then $L = 2$ and $\pi = u(1+i)$, where u is a unit in G .
- (D) Let $\delta = (1+i)$. Then δ and primes of the types q and α described above are, together with their associates, the only primes in G .

Exercise 5.10. Find the prime factors in G of the following “integers”:

12, 25, 19, 325, $6 + 7i$, $7 + 7i$, $3 + 5i$, and $-1 + 13i$. (You will need to look at norms.)

Exercise 5.11. Let α be in G and $\alpha \neq 0$. Then

$$\alpha = u\delta^k q_1^{n_1} q_2^{n_2} \dots q_r^{n_r} \pi_1^{m_1} \pi_2^{m_2} \dots \pi_s^{m_s},$$

where u is a unit, $\delta = (1 + i)$, each q_i is $\mathbb{Z}$ -prime congruent mod 4 to 3, and each of the π 's is a prime factor of a $\mathbb{Z}$ -prime, p , such that $p \equiv 1 \pmod{4}$. The nonzero members of G consist of all such α 's.

Exercise 5.12. Use the result of Exercise 5.11 to describe the set of all norms of members of G .

Exercise 5.13. Let n be in $\mathbb{Z}$. Then n is a sum of two squares in $\mathbb{Z}$ iff n is the norm of a member of G . What positive integers are sums of two squares?

Exercise 5.14. Decide whether each of these positive integers is the sum of two squares: 26, 27, 28, 29, 30, 31, 32, 325, 1200.

Exercise 5.15. If each of x and y is in $\mathbb{Z}$ and each is the sum of two squares, then so is xy .

Exercise 5.16. Let each of a , b , c , and d be in $\mathbb{Z}$. Exploit factorization in G to find x and y in $\mathbb{Z}$ such that

$$(a^2 + b^2)(c^2 + d^2) = x^2 + y^2.$$

At this point we do a few more exercises in G with no particular goal other than the fun you are going to have. That's not quite right; we will need some of the results, but the main goal is fun.

Exercise 5.17. Show that each of $1 + 3i$, 6 , $2 + 4i$, $3 + 5i$, and $3 - i$ is divisible in G by δ but that neither of $2 + 3i$, $5i$, $3 + 4i$, nor $5 + 6i$ is divisible by δ .

Exercise 5.18. Let $\alpha = a + bi$ be in G . Then $\alpha \equiv 0 \pmod{\delta}$ iff $a \equiv b \pmod{2}$, and $\alpha \equiv 1 \pmod{\delta}$ iff $a \not\equiv b \pmod{2}$, $G_\delta = \{[0], [1]\}$, and G_δ is a field that is isomorphic to $\mathbb{Z}_2$. This means there is a bijection between the two fields that is an isomorphism between the two additive groups and between the two multiplicative groups.

(We didn't define G_δ but we have met it before. To make your definition maybe you will need to go back and look at Ex. 3.7 to see how it is as natural to speak of G_δ or G_2 as to speak of $\mathbb{Z}_n$.)

Exercise 5.19. In G_δ , each $[a + bi]$ is either $[0]$ or $[1]$ by Exercise 5.18. Take a few members of G and decide to which class each belongs. Make operation tables for G_δ .

Definition. (This definition is due to Charles Yeomans. Charles had fun doing these exercises and made contributions to their evolution.) Let α be in G . Then α is “**even**” if α is divisible by δ (that is, of course, if $\alpha \equiv 0 \pmod{\delta}$) and α is “**odd**” if $\alpha \equiv 1 \pmod{\delta}$.

Exercise 5.20. The sum of two even members or of two odd members of G is even. The sum of an odd and an even member of G is odd. The product of an even member with any member is even, and the product of two odd members of G is odd.

Exercise 5.21. Let $\alpha = (2 + i)$. Then G_α has exactly five members. Can you identify them and make operation tables? Can you show that G_α is a field?

Exercise 5.22. G_3 has nine members. Can you identify them and make operation tables? Can you show that G_3 is a field?

Exercise 5.23. G_5 has 25 members. Can you identify them? Can you show that G_5 is not a field?

Exercise 5.24. How many members do you think G_9 has? How about G_{2+3i} ?

Exercise 5.25. What would it mean to say that $\phi_G(2 + i) = 4$? That $\phi_G(3) = 8$? Can you find $\phi_G(5)$? To what is $(1 + i)^4$ congruent mod $(2 + i)$? State Fermat’s Little Theorem for G and give Euler’s generalization.

5.2 Pythagorean Triples in $\mathbb{Z}$

Now we are going to do a few exercises dealing with **Pythagorean Triples** in $\mathbb{Z}$; that is, all triples $\{x, y, z\}$ of positive integers such that $x^2 + y^2 = z^2$. Equivalently, we are considering all right triangles with integers for side-lengths. Every carpenter knows one Pythagorean triple, $\{3, 4, 5\}$ (why does he know it?); some know another: $\{5, 12, 13\}$; these have been known since antiquity, not only by Greeks, but also by Chinese, Hindus, Arabs and probably others. You are soon going to know how to generate all infinitely many such triples. We are indebted to the Greeks (more specifically, to Diophantus) for the genesis of the method that you are about to learn, which evolved during a period of centuries in the minds and hands of Greeks, Hindus and Arabs, beginning around 300-200 B.C. You can probably sense that G is again a good setting in which to study the problem since we factor $x^2 + y^2$ in G . (We could confine our attention to $\mathbb{Z}$.)

Definition. A **Primitive Pythagorean Triple (PPT)** in $\mathbb{Z}$ is a Pythagorean Triple, $\{x, y, z\}$, in which x and y are relatively prime.

Exercise 5.26. If $\{x, y, z\}$ is a PPT in $\mathbb{Z}$, then the members of the triple are relatively prime in pairs, z is odd and exactly one of x and y is odd. (Consider $x^2 + y^2 = z^2 \pmod{4}$.) We arrange the nomenclature so that x is odd and regard the triple as an ordered triple (x, y, z) .

Exercise 5.27. Every Pythagorean Triple can be gotten from a PPT by multiplying the members through by some positive integer.

Exercise 5.28. Let (x, y, z) be a PPT in $\mathbb{Z}$. Then $x^2 + y^2 = z^2$ and

$$(x + iy)(x - iy) = z^2 \text{ in } G.$$

Exercise 5.29. The factors on the left side are both odd and they are coprime. (If π is a prime dividing each, then π divides their sum and their difference.)

Exercise 5.30. The Fundamental Theorem of Arithmetic in G implies that each factor on the left in Exercise 5.28 is a unit times a square. Thus, for some unit u and some odd number a in G , $x + iy = ua^2$.

Exercise 5.31. Let $a = t + si$, where of course t and s are in $\mathbb{Z}$, one even and the other odd. Then

$$ua^2 = u(t^2 - s^2 + 2its) = u(t^2 - s^2) + u(2ts)i = x + iy.$$

Exercise 5.32. Since the real part of ua^2 is x , and x is odd, u is neither i nor $-i$.

Exercise 5.33. If $u = 1$, then $x + iy = a^2$; if $u = -1$, $x + iy = (ia)^2$. In either case, $x + iy$ is a square; $x + iy = A^2 = (C + iD)^2 = (C^2 - D^2) + (2CD)i$. Here C and D are coprime, and one is even, the other odd.

Exercise 5.34. If (x, y, z) is a PPT in $\mathbb{Z}$, then there are positive integers, C and D , with $\gcd(C, D) = 1$, one even and the other odd, such that

$$x = C^2 - D^2, y = 2CD, \text{ and } z = C^2 + D^2.$$

Exercise 5.35. Find a pair (C, D) , guaranteed by Ex. 5.34, for each of the following PPT's: $(3, 4, 5)$, $(5, 12, 13)$, $(7, 24, 25)$ and $(15, 8, 17)$.

Exercise 5.36. Let M denote the set of all pairs, (C, D) , of coprime positive integers, C odd and D even. Let P denote the set of all PPT's in $\mathbb{Z}$. The map

$$(C, D) \rightarrow (|C^2 - D^2|, 2CD, C^2 + D^2),$$

is a bijection from M to P .

Now you know how to generate all PPT's, and you know that you won't waste time using different generators to get the same PPT.

Exercise 5.37. Generate some PPT's, using pairs from M . Find an infinite set of Pythagorean triples, using just one PPT to provide a start.

As I have mentioned before, Fermat's Last Theorem states that if $n > 2$, $x^n + y^n = z^n$ is not solvable in positive integers. Fermat thought he had a truly remarkable proof of the theorem, he did not communicate the proof, and it was a challenging problem for centuries. Fermat was almost certainly mistaken.

Of course, when one has solved the Pythagorean Triple problem, he thinks about integer solutions of $x^3 + y^3 = z^3$ and, more generally, of $x^n + y^n = z^n$. Let $\omega = \frac{-1+\sqrt{3}i}{2}$ and let $K = \{a + b\omega : a \text{ and } b \text{ are in } \mathbb{Z}\}$. K is an integral domain which includes the domain J of Section 1.3. Just as $x^2 + y^2$ factors into linear factors in the Gaussian integers, the polynomial, $x^3 + y^3$, factors into linear factors in K ; K "enjoys" unique factorization. One can use K to prove Fermat's ($n = 3$) Last Theorem. Similar techniques work for n a prime greater than 2 if the appropriate integral domain "enjoys" unique factorization; some don't. (To prove the theorem it suffices to prove it for $n = 4$, which is easy, and for n an odd prime. Why?) Despite many attempts to overcome the obstacles, this line of attack on the problem failed. The theorem was finally proved in the 1990's using powerful newly developed and diverse theories, which are beyond the scope of these exercises. The original version of the proof (by Andrew Wiles) filled 200 pages.

Index

- abelian, 6
- algebraically closed, 13
- arithmetic function, 29
- automorphism, 28
- casting out nines, 1
- Chinese Remainder Theorem, 27
- commutative, 6
- commutative ring, 6
- composite, 4
- congruence, 21
- coprime, 4
- cyclic group, 24
- degree, 12
- Dirichlet Product, 29
- divides, 1
- divisor theorem, 2
- Euclidean Algorithm, 3
- Euler ϕ -function, 24
- Euler's generalization
 - of Fermat's Little Theorem, 26
- even Gaussian Integer, 45
- Fermat number, 18
- Fermat prime, 18
- Fermat's Last Theorem, 10
- Fermat's Little Theorem, 26
- field, 11
- Fundamental Theorem of Algebra, 13
- Fundamental Theorem of Arithmetic, 5
- Gaussian integer, 6
- greatest common divisor, 2
- greatest integer function, 40
- group, 6
- Gauss's Lemma, 38
- homomorphism, 27
- ideal, 8
- integers, 1
- integral domain, 7
- isomorphism, 27
- Lagrange's Theorem, 26
- lattice point, 41
- least common multiple, 16
- Legendre Symbol, 36
- linear combination, 4
- Mersenne prime, 18
- Möbius Function, 30
- Möbius Inversion Formula, 31
- multiple, 1
- multiplicative function, 19
- norm, 7
- odd Gaussian integer, 45
- order of a group, 24
- order of a member of a group, 25
- Perfect Number, 19
- prime, 4
- Prime Number Theorem, 18
- Primitive Pythagorean triple, 45
- primitive root, 31
- principal ideal, 8
- Pythagorean triple, 45
- Quadratic Reciprocity Law, 36
- quadratic residue, 35
- Rational Root Theorem, 15
- relatively prime, 4
- repeating decimals, 33
- Riemann Zeta Function, 31
- ring, 1
- ring of integers (mod n), 23
- ring with unity, 6
- secret codes, 33
- squares (mod p), 35
- subgroup, 8

sum of two squares, 6
twin primes, 17
unit, 1
Wilson's Theorem, 36

Notes to the Instructor

¹The instructor might choose to talk about the notion of “Euclidean domains” in connection with this failure of unique factorization in J .

²We shall later, of course, see the finite fields $\mathbb{Z}_p$ where p is a prime. But it is interesting to note that at the very end of the Notes we naturally encounter a few finite fields that have p^2 elements in them, as in Ex. 5.22.

³In Ex. 1.98 it might be pointed out that we are here discovering the three cube roots of unity – something that will be new to some students.

⁴In Ex. 2.2 one might comment that after replacing each of the integers a and b by an appropriate product of primes, we see that the equation $p(b^n) = a^n$ cannot hold, for the prime p cannot occur the same number of times on both sides of this equation if $n > 1$.

⁵Ex. 2.9 outlines Euclid’s ancient algorithm for finding a new prime, given any finite set of primes: Just find a prime factor of one plus the product of all the primes in the set. It might be pointed out, however, that Euclid’s method – though constructive – is hopelessly inefficient. Anyone who knows the first 25 primes and wants another prime would not want to multiply the 25 known primes together and attempt to factor the product plus one.

⁶In Ex. 2.12 it has been my experience that students generally don’t work hard enough to find a counterexample to the natural conjecture here. In this connection a problem related to the famous Euler polynomial $n^2 - n + 41$ might be helpful. This polynomial takes prime values for $n = 1, 2, 3, \dots, 40$, but this fact doesn’t mean you can’t find a non-prime output ($n = 41$).

⁷After the work based on the fact that 10 is congruent to 1 mod 9, one might assign a problem based on the fact that 10 is congruent to -1 mod 11. Thus, a number in decimal notation is divisible by 11 iff the alternating sum of its divisors is.

⁸Cross gives as an example the message, “I passed my comps.” This refers to his students’ passing their comprehensive examination in mathematics.

⁹Perhaps the instructor should make up some more complicated quadratic reciprocity problems for the students to work out, by introducing quadratic equations modulo larger primes than are considered here.

¹⁰In connection with Exercise 4.19 it might not be a good idea to interrupt the flow of ideas here, but the result of this exercise quickly shows that there are infinitely many primes of the form $4n + 1$: For any prime factor of $[N!]^2 + 1$ is seen to be such a prime p by this exercise and it is easy to see that $p > N$. (An easier argument based upon a simple

modification of Euclid's familiar argument shows that there are also infinitely many primes of the form $4n + 3$. Proofs of these relatively simple special cases of Dirichlet's Theorem might make good assignments for students.

Additional materials in support of this course are available. Interested instructors are invited to correspond with W. M. Priestley at wpriestl@sewanee.edu.